

9

Laser cooling and trapping

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¹The inhomogeneous magnetic field deflects the atoms in the Stern–Gerlach experiment but has a negligible effect on the speed.

²This condition does not hold true for cold atoms moving through a standing wave of light where the intensity varies significantly over short distances (comparable with the optical wavelength, see Section 9.7).

³Chu and co-workers also developed a physically equivalent description.

In previous chapters we have seen how laser spectroscopy gives Doppler-free spectra and also how other older techniques of radio-frequency and microwave spectroscopy can resolve small splittings, e.g. hyperfine structure. These methods just observe the atoms as they go past,¹ but this chapter describes the experimental techniques that use the force exerted by laser light to slow the atomic motion and manipulate atoms. These techniques have become extremely important in atomic physics and have many applications, e.g. they have greatly improved the stability of the caesium atomic clocks that are used as primary standards of time around the world. We shall look at the forces that laser light exerts on an atom in some detail since this aspect contains most of the atomic physics. In many of the cases studied in this chapter, the atom's motion follows straightforwardly from Newton's laws once the force is known—an atom behaves like a classical particle, localised at a particular point in space, when the atomic wavepacket has a spread which is small compared to the distance over which the potential energy varies.²

The first laser cooling experiments were carried out on ions that were trapped by electric fields and then cooled by laser radiation. In contrast, it is difficult to confine atoms at room temperature, or above, because of the smaller electromagnetic forces on neutral particles. Therefore the pioneering experiments used light forces to slow atoms in an atomic beam and then confined the cold atoms with a magnetic field. The great success of laser cooling led to the award of the 1997 Nobel prize in physics to Steven Chu, Claude Cohen-Tannoudji and William Phillips. To describe the development of the subject we consider their contributions in the following order. We start from an explanation of the light force on atoms in terms of the scattering of photons. The research group of Phillips used this force to slow an atomic beam (Section 9.2). Chu and co-workers then demonstrated the method known as the optical molasses technique, that cools the motion of atoms in all three dimensions to give a very cold atomic vapour (Section 9.3). This led directly to the development of the so-called magneto-optical trap (Section 9.4) used in the majority of atom-trapping experiments today.

The interaction of the atoms with the light field turned out to be much more subtle than first supposed, and experiments showed that the optical molasses technique produced even lower temperatures than predicted. Cohen-Tannoudji and Jean Dalibard explained this behaviour by a new mechanism called Sisyphus cooling.³ This mechanism is described towards the end of the chapter (Section 9.7) since it does not

fall neatly into either category of the radiation force, i.e. (a) the scattering force that arises from absorption of light and spontaneous emission, and (b) the dipole force, described in Section 9.6. The forces on microscopic particles have properties similar to the forces on individual atoms, and this analogy is used to introduce the dipole force in Section 9.5. Atoms that have been pre-cooled by the other laser cooling techniques described in this chapter can be cooled even further using Raman transitions (Section 9.8). Finally, Section 9.9 describes an atomic fountain which requires laser-cooled atoms.

9.1 The scattering force

The idea that radiation has momentum (and energy) goes back to James Clerk Maxwell in the nineteenth century. It follows from the conservation of momentum that when an object absorbs radiation its momentum changes. The force on the object equals the rate of change of momentum. Therefore the force equals the rate at which the light delivers momentum—this is the same as the rate at which the light delivers energy divided by the speed of light.⁴ Therefore radiation of intensity I exerts a force on area A given by

$$F_{\text{rad}} = \frac{IA}{c}. \quad (9.1)$$

Equivalently, the radiation pressure is $F_{\text{rad}}/A = I/c$. The quantity IA equals the power absorbed, e.g. for $IA = 1 \text{ W}$ the force is $F = 3.3 \times 10^{-9} \text{ N}$. At a surface that reflects the radiation back on itself the momentum change is twice as large and gives twice the force in eqn 9.1. Although small, the radiation force has observable effects in astrophysics, e.g. the outward radiation pressure balances gravity in stars, and the tails of comets point away from the sun (rather than trailing behind as for shooting stars in the atmosphere). Note, however, that although radiation pressure does have some effect on the dust and ice particles that form the tail of a comet, the solar wind is also important—the stream of particles emanating from the sun hit the particles in the comet tail and even the relatively low pressure in space leads to a force comparable to that from radiation pressure.⁵ Radiation forces have a dramatic effect on atoms because the peak absorption cross-section $\sigma_{\text{abs}}(\omega_0)$ is much greater than the physical size of the atom (see eqn 7.81).⁶

Lasers produce well-collimated monochromatic beams of light that can slow atoms in an atomic beam, as illustrated in Fig. 9.1. A counter-propagating laser beam exerts a force of $F = -\sigma_{\text{abs}}I/c$ on an atom, where the minus sign indicates a force in the opposite direction to the motion. This expression in terms of the absorption cross-section shows that the light does not have to be considered as quantised in order to calculate the force, but it is convenient to describe the processes in terms of photons. Each absorbed photon gives the atom a kick in the direction opposite to its motion and spontaneously-emitted photons go in all directions, so that the scattering of many photons gives an average

⁴The energy of radiation divided by its momentum equals c . For a photon

$$\frac{\text{energy}}{\text{momentum}} = \frac{\hbar\omega}{\hbar k} = \frac{\omega}{k} = c.$$

Of course, Maxwell showed this by classical electromagnetism, not in terms of photons. (The ratio does not depend on $\hbar$.) This and other aspects of the pressure due to electromagnetic radiation are discussed in Bleaney and Bleaney (1976, Section 8.8).

⁵The radiation from the sun has an intensity of 1.4 kW m^{-2} at the Earth. Thus the radiation pressure at the Earth's orbit is $5 \times 10^{-6} \text{ N m}^{-2}$, or slightly less than $\sim 10^{-10}$ times atmospheric pressure.

⁶Alkali atoms in a vapour have a large fraction of their absorption strength concentrated in a narrow range centred at the frequency of the resonance line.

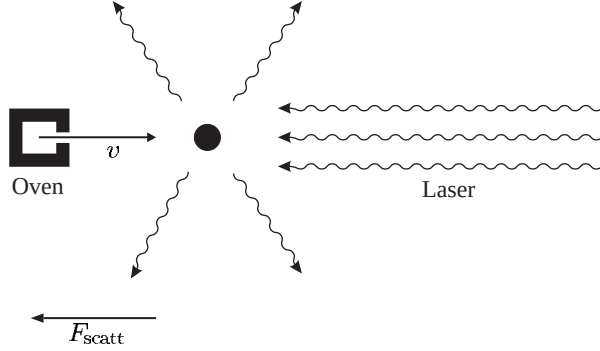


Fig. 9.1 For an atom moving towards the laser, each absorbed photon gives the atom a kick in the direction opposite to its motion and the scattered photons go in random directions, resulting in a force that slows the atom.

force that slows the atom down. The magnitude of this scattering force equals the rate at which the absorbed photons impart momentum to the atom:

$$F_{\text{scatt}} = (\text{photon momentum}) \times (\text{scattering rate}). \quad (9.2)$$

The scattering rate is $R_{\text{scatt}} = \Gamma \rho_{22}$, and ρ_{22} , the fraction of the population in level 2, is given in eqn 7.69, so that

$$R_{\text{scatt}} = \frac{\Gamma}{2} \frac{\Omega^2/2}{\delta^2 + \Omega^2/2 + \Gamma^2/4}. \quad (9.3)$$

The frequency *detuning* from resonance $\delta = \omega - \omega_0 + kv$ equals the difference between the laser frequency ω and the atomic resonance frequency ω_0 taking into account the Doppler shift kv . The Rabi frequency and saturation intensity are related by $I/I_{\text{sat}} = 2\Omega^2/\Gamma^2$ (see eqn 7.86)⁷ and photons have momentum $\hbar k$, so that⁸

$$F_{\text{scatt}} = \hbar k \frac{\Gamma}{2} \frac{I/I_{\text{sat}}}{1 + I/I_{\text{sat}} + 4\delta^2/\Gamma^2}. \quad (9.4)$$

As $I \rightarrow \infty$ the force tends to a limiting value of $F_{\text{max}} = \hbar k \Gamma/2$. The rate of spontaneous emission from two-level atoms tends to $\Gamma/2$ at high intensities because the populations in the upper and lower levels both approach 1/2. This follows from Einstein's equations for radiation interacting with a two-level atom that has degeneracy factors $g_1 = g_2 = 1$.

For an atom of mass M this radiation force produces a maximum acceleration that we can write in various forms as

$$a_{\text{max}} = \frac{F_{\text{max}}}{M} = \frac{\hbar k \Gamma}{M 2} = \frac{v_r}{2\tau}, \quad (9.5)$$

where τ is the lifetime of the excited state. The **recoil velocity** v_r is the change in the atom's velocity for absorption, or emission, of a photon at wavelength λ ; it equals the photon momentum divided by the atomic mass: $v_r = \hbar k/M \equiv h/(\lambda M)$. For a sodium atom $a_{\text{max}} = 9 \times 10^5 \text{ m s}^{-2}$, which is 10^5 times the gravitational acceleration. For the situation shown in Fig. 9.1 the atom decelerates at a rate

$$\frac{dv}{dt} = v \frac{dv}{dx} = -a, \quad (9.6)$$

⁷Generally speaking, intensity is more directly related to experimental parameters than the Rabi frequency, but we shall use both I and Ω in this chapter. As noted previously, there are other definitions of the saturation intensity in common use that differ by a factor of 2 from the one used here.

⁸This statement relies on the comprehensive description of two-level atoms interacting with radiation given in Chapter 7.

where a is positive. For constant deceleration from an initial velocity v_0 at $z = 0$, integration gives the velocity as a function of distance:

$$v_0^2 - v^2 = 2az. \quad (9.7)$$

Typically, the deceleration is half the maximum value $a = a_{\max}/2$ to ensure that atoms are not left behind.⁹ Hence the stopping distance is

$$L_0 = \frac{v_0^2}{a_{\max}}. \quad (9.8)$$

A typical apparatus for slowing a sodium atom ($M \simeq 23$ a.m.u.) uses the parameters in the following table, where the initial velocity v_0 is taken to be the most probable velocity in a beam (given in Table 8.1).

Most probable velocity in beam ($T = 900$ K)	v_0	1000 m s^{-1}
Resonance wavelength	λ	589 nm
Lifetime of the excited state	τ	16 ns
Recoil velocity	$v_r = h/(\lambda M)$	3 cm s^{-1}
Stopping distance (at half of the maximum deceleration)	$2v_0^2\tau/v_r$	1.1 m

A distance of 1 m is a convenient length for an experiment, and all alkali metals require surprisingly similar stopping distances. Although the heavier elements have a lower deceleration, they also have lower initial velocities because of their higher mass and the lower temperature required to give sufficient vapour pressure in an oven for a high-flux atomic beam, e.g. eqn 9.8 gives $L_0 = 1.2 \text{ m}$ for rubidium and a most probable velocity of $v_0 = v_{\text{beam}} = 360 \text{ m s}^{-1}$ at $T = 450$ K (from the data in Table 9.1).

The majority of laser cooling experiments have been carried out with sodium or rubidium.¹⁰

The calculation of the stopping distance assumes a constant deceleration, but for a given laser frequency atoms only experience a strong force over a narrow range of velocities, $\Delta v \sim \Gamma/k$, for which the atoms have a

⁹Fluctuations of the force about its average value arise from the randomness in the number of photons scattered per unit time, as described in Section 9.3.1.

¹⁰Other elements such as magnesium (an alkaline earth metal) have ultraviolet transitions with shorter lifetimes that scatter photons of higher momentum. Thus the atoms stop in a shorter distance; however, it is technically more difficult to obtain continuous-wave ultraviolet radiation.

Table 9.1 Properties of some elements used in laser cooling experiments.

Element	Atomic mass number	Wavelength of resonance (nm)	Lifetime of the excited state (ns)
H	1	121.6	1.6
Li	7	671	27
Na	23	589	16
K	39	767	26
Rb	85, 87	780	27
Cs	133	852	31

¹¹This range is similar to the width of the hole burnt in the ground-state population in saturated absorption spectroscopy. In practice, power broadening makes the homogeneous width larger than the natural width.

range of Doppler shifts approximately equal to the homogeneous width of the transition.¹¹ Atoms that interact strongly with the laser light slow down until the change in their Doppler shift takes them out of resonance with the light. This change must be compensated for in order to keep the force close to its maximum value throughout the slowing process.

9.2 Slowing an atomic beam

The two pioneering laser cooling experiments used different methods to compensate for the change in Doppler shift as the atoms slowed down. William Phillips and co-workers used the ingenious method shown in Fig. 9.2 in which the atomic beam travels along the axis of a tapered solenoid; the Zeeman effect of the varying magnetic field perturbs the atomic energy levels so that the transition frequency matches a constant laser frequency. In the other method the laser frequency was changed and this so-called chirp cooling is described in the next section. From eqns 9.7 and 9.8 we see that during constant deceleration the velocity at distance z from the starting point is given by

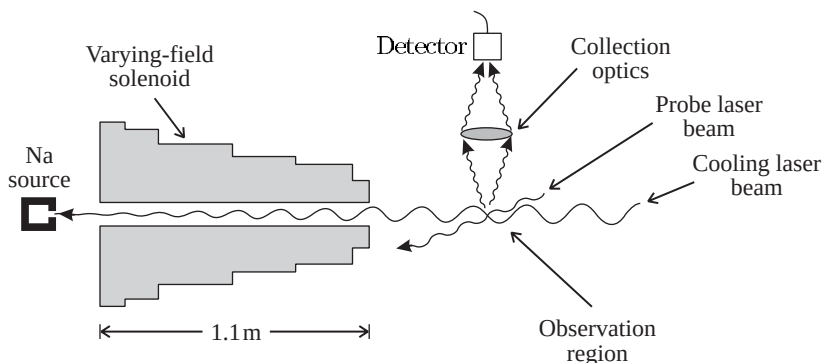
$$v = v_0 \left(1 - \frac{z}{L_0}\right)^{1/2}. \quad (9.9)$$

To compensate for the change in Doppler shift as the atoms slow down from v_0 to the chosen final velocity, the frequency shift caused by the Zeeman effect needs to obey the condition

$$\omega_0 + \frac{\mu_B B(z)}{\hbar} = \omega + kv. \quad (9.10)$$

On the left-hand side, the Zeeman shift for an atomic magnetic moment μ_B increases the atomic resonance frequency from ω_0 , its value at zero

Fig. 9.2 The first Zeeman slowing experiment. The solenoid produces a magnetic field that varies with position along the atomic beam so that the Zeeman shift compensates for the change in the Doppler shift as atoms decelerate. A probe laser beam intersects the slow atomic beam at a point downstream and the laser frequency is scanned to give a fluorescent signal proportional to the velocity distribution, similar to that shown in Fig. 9.4 (from a different experiment). This procedure records the component of the atomic velocity along the probe beam and the angle of intersection must *not* be 90° . (Here we have the opposite requirement to that in Fig. 8.2.) Adapted from Phillips *et al.* (1985).



field; on the other side of the equation, the Doppler shift adds to the laser frequency ω . Hence we find from eqn 9.9 that the required magnetic field profile is

$$B(z) = B_0 \left(1 - \frac{z}{L_0}\right)^{1/2} + B_{\text{bias}} \quad (9.11)$$

for $0 \leq z \leq L_0$, where

$$B_0 = \frac{\hbar v_0}{\lambda \mu_B}. \quad (9.12)$$

If $\mu_B B_{\text{bias}} \simeq \hbar\omega - \hbar\omega_0$ then the atoms come to a complete stop at the end of the tapered solenoid; generally, it is more useful to leave the atoms with a small velocity so that they travel out of the tapered solenoid to a region where experiments, or further cooling, can be performed. Figure 9.3(a) shows the field profile for $\omega \simeq \omega_0$ and $B_{\text{bias}} \simeq 0$, so that the maximum field at the entrance to the solenoid is about B_0 ; Fig. 9.3(b) shows the field profile for a different choice of B_{bias} that requires a lower magnitude of the field.

Example 9.1 For a transition between a state with quantum numbers F and M_F and an excited state with quantum numbers F' and $M_{F'}$, the Zeeman effect causes an (angular) frequency shift of $(g_{F'} M_{F'} -$

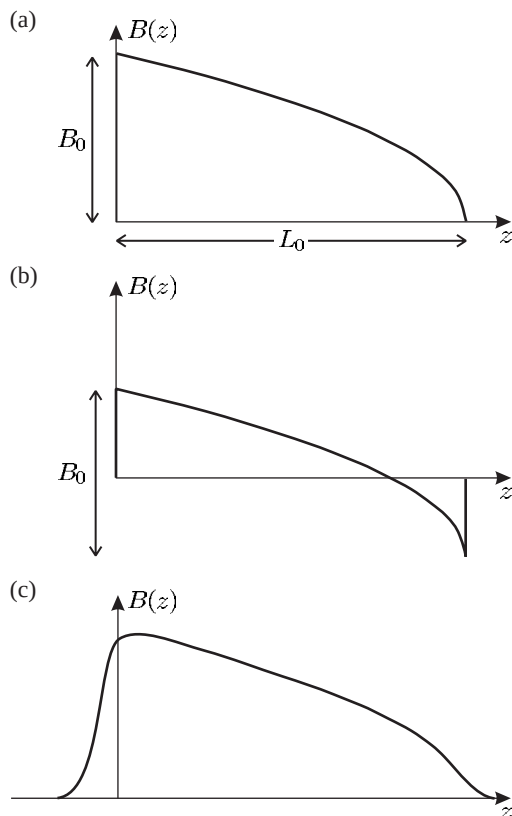


Fig. 9.3 The magnetic field along the tapered solenoid in the first Zeeman slowing experiment (see Fig. 9.2) varied with position as shown in (a). This magnetic field is described by eqn 9.11. Nowadays, some experiments use the variant shown in (b) where the field drops to zero and then reverses; this gives the same decrease in velocity for a given change in field B_0 , but has the following three advantages: (i) the field has a lower maximum value so that the coils need less current-turns; (ii) the ‘reverse slower’ produces less field at positions downstream, $z > L_0$, because the contributions from the coils with currents in opposite directions tend to cancel out; and (iii) the abrupt change in the field at the exit helps the atoms to leave the solenoid cleanly (the atoms see a sudden increase in the frequency detuning of the light from resonance that cuts off the radiation force). In a real solenoid the field changes gradually, as illustrated in (c). This smoothing has little influence on the overall length required because it does not significantly affect the early part of the slowing process.

¹²Other alkalis have corresponding transitions between ‘fully-stretched’ states for which F has the maximum value for a given I and J ; and $M_F = F$ or $M_F = -F$. Sodium has a nuclear spin of $I = 3/2$.

¹³From eqn 9.10 we see that the radiation exerts the strongest force on atoms with a velocity v_f at the end of the solenoid that is given by

$$kv_f \simeq \omega_0 + \frac{\mu_B B_{\text{bias}}}{\hbar} - \omega. \quad (9.14)$$

The actual final velocity will be lower than v_f because the atoms remain in the laser light after they have emerged from the solenoid and are slowed further. A rough estimate suggests that the atomic velocity is lower than v_f by an amount corresponding to a Doppler shift of several line widths (e.g. $3\Gamma/k$), but this depends on the interaction time with the light (which itself depends on the velocity and distance travelled). It is difficult to obtain very low final velocities without stopping the atoms completely and pushing them back into the solenoid.

$g_F M_F) \mu_B B / \hbar$. For the $3p\ ^2P_{3/2} - 3s\ ^2S_{1/2}$ line in sodium, the transition between the hyperfine levels $F' = 3$, $M_{F'} = 3$ and $F = 2$, $M_F = 2$ has $g_{F'} M_{F'} - g_F M_F = 1$, so its Zeeman shift is as assumed in eqn 9.10. This transition leads to a closed cycle of absorption and spontaneous emission because selection rules dictate that the excited state can only decay back to the initial state. This transition was used in the first slowing experiment shown in Fig. 9.2.¹² Hence for $v_0 = 1000\text{ m s}^{-1}$ we find from eqn 9.12 that

$$B_0 = 0.12\text{ T}, \quad (9.13)$$

which is well within the capability of standard magnet coils. In this magnetic field a sodium atom has a Zeeman shift equal to the Doppler shift of $\Delta f = v_0/\lambda = 1.7\text{ GHz}$ (cf. the natural width $\Gamma/2\pi = 10\text{ MHz}$).

The important feature of the Zeeman slowing technique is that it reduces the velocity of a large fraction of the atoms in a beam to a low final value v_f . Any atoms that start with velocities within the range v_0 to v_f interact with the laser radiation at some position along the solenoid and are swept along in the slowing process. The calculations in this section show how this works in principle, but the equations do not give the value of the final velocity v_f accurately for the following reason. The stopping distance is proportional to the square of the initial velocity (eqn 9.8), so during deceleration from $v_0 = 100\text{ m s}^{-1}$ to $v_f = 0\text{ m s}^{-1}$ the atoms only move 1 cm; and deceleration from $v_0 = 33\text{ m s}^{-1}$ to zero occurs within 1 mm. Thus the final velocity depends critically on what happens at the end of the solenoid and in the fringing field that extends beyond.¹³ In practice, the laser frequency is adjusted so that the atoms have sufficient velocity to continue along the apparatus. Various methods for extracting the atoms have been developed, such as that shown in Fig. 9.3(b) where the field changes abruptly at the end of the solenoid so the interaction with the light shuts off cleanly.

9.2.1 Chirp cooling

In the other pioneering experiment to laser cool a sodium beam, the laser frequency was changed to keep track of the Doppler shift as the atoms slowed down. This method has become known as chirp cooling—a chirped pulse is one in which the frequency sweeps rapidly. This name derives from an analogy with bird-song, where the pitch of the sound changes rapidly. We can calculate the sweep time from the number of photon kicks required to stop the atom: $\mathcal{N} = v_0/v_r$. An atom scatters photons at a maximum rate of $\Gamma/2 = 1/(2\tau)$. Therefore $\mathcal{N}$ photons are scattered in a time of $2\mathcal{N}\tau$, or at half the maximum deceleration it takes twice as long, i.e. $4\mathcal{N}\tau$. For a beam of sodium atoms with parameters given in Example 9.1, $\mathcal{N} = 34\,000$ and the sweep time is $4\mathcal{N}\tau = 2 \times 10^{-3}\text{ s}$. The frequency of the light must be swept over a range of more than 1 GHz in a few milliseconds. Tuneable dye lasers cannot scan this quickly and so the experimenters used electro-optic modulators and radio-frequency techniques to change the frequency of

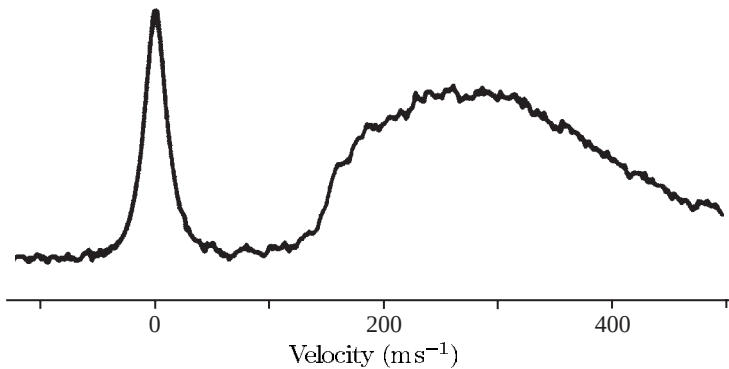


Fig. 9.4 The atomic velocity distribution produced by chirp cooling caesium atoms with radiation from semiconductor diode lasers. The trace shows the experimentally observed fluorescence from the atoms as the laser frequency was scanned over a frequency range greater than the initial Doppler shift of the atoms in the atomic beam. From Steane (1991).

the light. Nowadays, chirp cooling of heavy alkalis such as rubidium and caesium can be carried out by directly scanning the frequency of infra-red semiconductor diode lasers—Fig. 9.4 shows the results of such an experiment. It can be seen that the laser cooling sweeps atoms to lower velocities to produce a narrow peak in the velocity distribution. It is the spread of the velocities within this peak that determines the final temperature, not the mean velocity of these atoms. The atoms have a much smaller spread of velocities than at room temperature so they are cold.

9.3 The optical molasses technique

In an atomic beam the collimation selects atoms moving in one direction that can be slowed with a single laser beam. Atoms in a gas move in all directions and to reduce their temperature requires laser cooling in all three directions by the configuration of three orthogonal standing waves shown in Fig. 9.5—the light along the Cartesian axes comes from the same laser and has the same frequency. At first, you might think that this symmetrical arrangement has no effect on an atom since there are equal and opposite forces on an atom. However, the radiation forces from the laser beams balance each other only for a stationary atom, which is what we want to achieve. For a moving atom the Doppler effect leads to an imbalance in the forces. Figure 9.5(b) shows the situation for a two-level atom in a pair of counter-propagating beams from a laser with a frequency below the atomic resonance frequency (red frequency detuning). Consider what happens in the reference frame of an atom moving towards the right, as shown in Fig. 9.5(c). In this frame the Doppler effect leads to an increase in the frequency of the laser beam propagating in the direction opposite to the atom's velocity. This Doppler shift brings the light closer to resonance with the atom and thereby increases the rate of absorption from this beam. This leads to a resultant force that slows the atom down.¹⁴ Expressed mathematically, the difference between the force to the right and that to the left is

¹⁴A similar situation arises for movement in any direction.

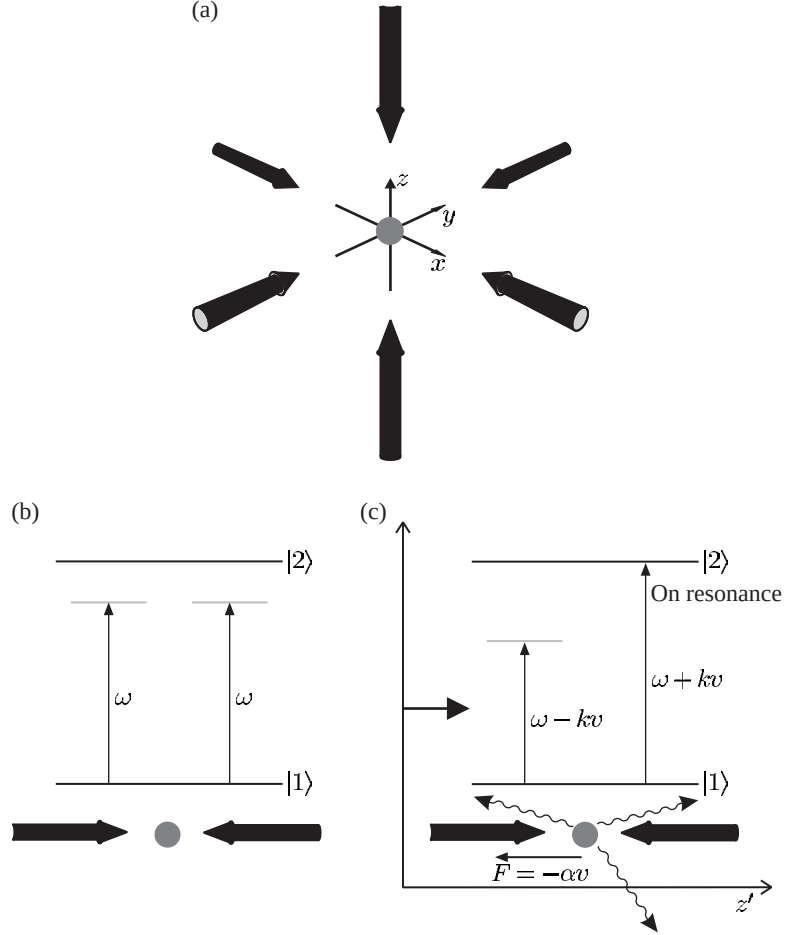


Fig. 9.5 ‘Optical molasses’ is the name given to the laser cooling technique that uses the configuration of three orthogonal pairs of counter-propagating laser beams along the Cartesian axes shown in (a). The laser beams are derived from the same laser and have a frequency ω that is slightly below the transition frequency between the two atomic levels 1 and 2. (b) A stationary atom in a pair of counter-propagating laser beams experiences no resultant force because the scattering is the same for each laser beam, but for a moving atom, as in (c), the Doppler effect leads to more scattering of the light propagating in the direction opposite to the atom’s velocity. (Part (c) is drawn in the rest frame of an atom moving at velocity v .) The imbalance in the forces occurs for all directions and damps the atomic motion.

$$\begin{aligned}
 F_{\text{molasses}} &= F_{\text{scatt}}(\omega - \omega_0 - kv) - F_{\text{scatt}}(\omega - \omega_0 + kv) \\
 &\simeq F_{\text{scatt}}(\omega - \omega_0) - kv \frac{\partial F}{\partial \omega} - \left[F_{\text{scatt}}(\omega - \omega_0) + kv \frac{\partial F}{\partial \omega} \right] \\
 &\simeq -2 \frac{\partial F}{\partial \omega} kv.
 \end{aligned} \tag{9.15}$$

Low velocities, $kv \ll \Gamma$, have been assumed. This imbalance in the forces arising from the Doppler shift can be written as

$$F_{\text{molasses}} = -\alpha v. \tag{9.16}$$

¹⁵Strictly,

$$F_{\text{scatt}} = \hbar k R_{\text{scatt}} \equiv \hbar \frac{\omega}{c} R_{\text{scatt}},$$

so

$$\frac{\partial F}{\partial \omega} = \frac{\hbar}{c} \left(R_{\text{scatt}} + \omega \frac{\partial R_{\text{scatt}}}{\partial \omega} \right),$$

but typically the second term is about $\omega/\Gamma \simeq 10^8$ larger than the first term.

The light exerts a frictional, or damping, force on the atom just like that on a particle in a viscous fluid. This analogy led the Americans who first demonstrated the effect (Chu *et al.* 1985) to call it the optical molasses technique (like treacle, or honey)—a name that seems to have stuck! Differentiation of eqn 9.4 gives the damping coefficient as¹⁵

$$\alpha = 2k \frac{\partial F}{\partial \omega} = 4\hbar k^2 \frac{I}{I_{\text{sat}}} \frac{-2\delta/\Gamma}{[1 + (2\delta/\Gamma)^2]^2}. \tag{9.17}$$

The term $I/I_{\text{sat}} \ll 1$ has been neglected in the denominator because this simple treatment of the optical molasses technique is only valid for intensities well below saturation where the force from each beam acts independently.¹⁶ Damping requires a positive value of α and hence $\delta = \omega - \omega_0 < 0$, i.e. a red frequency detuning (in accordance with the physical explanation of the optical molasses technique given above). For this condition the plots of the force in Fig. 9.6 have a negative gradient $\partial F/\partial v < 0$ at $v = 0$.

The above discussion of the optical molasses technique applies to one of pair of a counter-propagating laser beams. For the beams parallel to the z -axis, Newton's second law gives

$$\frac{d}{dt} \left(\frac{1}{2} M v_z^2 \right) = M v_z \frac{dv_z}{dt} = v_z F_{\text{molasses}} = -\alpha v_z^2. \quad (9.18)$$

The components of the velocity along the x - and y -directions obey similar equations, so that in the region where the three orthogonal pairs of laser beams intersect the kinetic energy $E = \frac{1}{2} M (v_x^2 + v_y^2 + v_z^2)$ decreases:

$$\frac{dE}{dt} = -\frac{2\alpha}{M} E = -\frac{E}{\tau_{\text{damp}}}. \quad (9.19)$$

Under optimum conditions the damping time $\tau_{\text{damp}} = M/(2\alpha)$ is a few microseconds (see Exercises 9.7 and 9.8). This gives the time-scale for the initial cooling of atoms that enter the laser beams with velocities within the capture range of the optical molasses technique, i.e. velocities for which the force has a significant value in Fig. 9.6. Equation 9.19 gives the physically unrealistic prediction that energy tends to zero because we have not taken into account the heating from fluctuations in the force.

¹⁶Saturation in two counter-propagating beams could be taken into account by the replacement $I/I_{\text{sat}} \rightarrow 2I/I_{\text{sat}}$ in the denominator of the expression for α (and also R_{scatt}), although if I/I_{sat} is not negligible a simple rate equation treatment is not accurate. We will see that for real atoms, as opposed to theoretical two-level atoms, the light field needs to be considered as a standing wave even for low intensities.

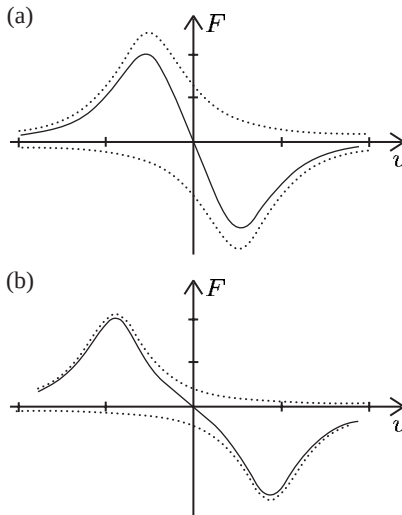


Fig. 9.6 The force as a function of the velocity in the optical molasses technique (solid lines) for (a) $\delta = -\Gamma/2$, and (b) $\delta = -\Gamma$. The damping is proportional to the slope of the force curve at $v = 0$. Note that the force is negative for $v > 0$ and positive for $v < 0$, so the force decelerates atoms. The forces produced by each of the laser beams separately are shown as dotted lines—these curves have a Lorentzian line shape and they are drawn with an FWHM of Γ appropriate for low intensities. For $\delta = 0$ (not shown in the figure) the forces from the two laser beams cancel each other for all velocities. The velocity capture range is approximately $\pm\Gamma/k$.

9.3.1 The Doppler cooling limit

We can write the force from a single laser beam as

$$\mathbf{F} = \mathbf{F}_{\text{abs}} + \delta\mathbf{F}_{\text{abs}} + \mathbf{F}_{\text{spont}} + \delta\mathbf{F}_{\text{spont}}. \quad (9.20)$$

The average of the force from the absorption of photons is the scattering force that we have already derived, $\overline{\mathbf{F}}_{\text{abs}} = \mathbf{F}_{\text{scatt}}$, and the random kicks from spontaneously-emitted photons average to zero, i.e. $\overline{\mathbf{F}}_{\text{spont}} = 0$. What we have not considered previously is the effect of the fluctuations in these two processes, i.e. $\delta\mathbf{F}_{\text{spont}}$ and $\delta\mathbf{F}_{\text{abs}}$.

The spontaneous emission that always accompanies $\mathbf{F}_{\text{scatt}}$ causes the atom to recoil in random directions. These recoil kicks lead to a random walk of the velocity, as shown in Fig. 9.7 (analogous to the Brownian motion of microscopic particles in air). A random walk of $\mathcal{N}$ steps gives a mean displacement proportional to $\sqrt{\mathcal{N}}$, or equivalently the mean square displacement equals $\mathcal{N}$ times the square of the step length. During a time t an atom scatters a mean number of photons

$$\mathcal{N} = R_{\text{scatt}}t. \quad (9.21)$$

Spontaneous emission causes the mean square velocity to increase as $v^2 = R_{\text{scatt}}t \times v_r^2$, or along the z -axis

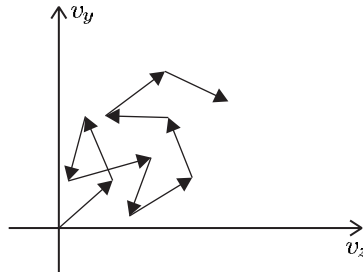
$$(\overline{v_z^2})_{\text{spont}} = \eta v_r^2 R_{\text{scatt}}t. \quad (9.22)$$

Each spontaneous photon gives a recoil kick in the z -direction of $\hbar k \cos \theta$ and the factor $\eta = \langle \cos^2 \theta \rangle$ is the angular average, e.g. for isotropic spontaneous emission $\eta = 1/3$.¹⁷

The fluctuations $\delta\mathbf{F}_{\text{abs}}$ arise because the atom does not always absorb the same number of photons in a time period t . Each absorption is followed by spontaneous emission and the mean number of such events in time t is given by eqn 9.21. Assuming that the scattering obeys Poissonian statistics, the fluctuations about the mean have a standard deviation of $\sqrt{\mathcal{N}}$ and cause a random walk of the velocity along the laser beam, on top of the change in velocity (acceleration or deceleration) caused by the mean force. This one-dimensional random walk caused by the fluctuations $\delta\mathbf{F}_{\text{abs}}$ leads to an increase in the velocity spread similar

¹⁷The spontaneously-emitted photons that go in the x - and y -directions lead to heating in those directions, that must be taken into account in a full three-dimensional treatment. For isotropic spontaneous emission the heating would be the same in all directions, i.e. $\overline{v_x^2}$ and $\overline{v_y^2}$ would increase at the same rate as $\overline{v_z^2}$. Radiation from an electric dipole oscillator is not isotropic, but this turns out not to be important for reasons discussed below.

Fig. 9.7 The recoil of an atom from each spontaneous emission causes the atomic velocity to change by the recoil velocity in a random direction. Thus the atom undergoes a random walk in velocity space with steps of length v_r . The equilibrium temperature is determined by the balance between this diffusive heating and the cooling. (For simplicity, only two directions are shown here.)



to that in eqn 9.22, but without the factor η (because all the absorbed photons have the same direction):

$$(\overline{v_z^2})_{\text{abs}} = v_r^2 R_{\text{scatt}} t. \quad (9.23)$$

This gives the effect of $\overline{\delta \mathbf{F}}_{\text{abs}}$ for a single laser beam. For an atom in two counter-propagating beams the radiation forces of the two beams tend to cancel (as in eqn 9.5) but the effect of the fluctuations is cumulative. The atom has an equal probability of absorbing photons from either beam (incident from the left and right, say) so it receives impulses that are randomly either to the left, or to the right, leading to a random walk of velocity along the beams.¹⁸

From eqns 9.22 and 9.23 we can find the heating arising from $\overline{\delta \mathbf{F}}_{\text{spont}}$ and $\overline{\delta \mathbf{F}}_{\text{abs}}$, respectively. Inserting these terms into eqn 9.18, and assuming that for a pair of beams the scattering rate is $2R_{\text{scatt}}$ (twice the rate for a single beam of intensity I), we find

$$\frac{1}{2} M \frac{d\overline{v_z^2}}{dt} = (1 + \eta) E_r (2R_{\text{scatt}}) - \alpha \overline{v_z^2}, \quad (9.24)$$

where

$$E_r = \frac{1}{2} M v_r^2 \quad (9.25)$$

is the recoil energy. Equation 9.24 describes the balance between heating and damping for an atom in a pair of counter-propagating beams, but in the optical molasses technique there are usually three orthogonal pairs of laser beams, as shown in Fig. 9.5. To estimate the heating in this configuration of six laser beams we shall assume that, in the region where the beams intersect, an atom scatters photons six times faster than in a single beam (this neglects any saturation). If the light field is symmetrical¹⁹ then the spontaneous emission is isotropic. Thus averaging over angles gives $\eta = 1/3$, but the overall contribution from $\overline{\delta \mathbf{F}}_{\text{spont}}$ is three times greater than for a pair of the laser beams. Therefore the factor $1 + \eta$ in eqn 9.24 becomes $1 + 3\eta = 2$ for the three-dimensional configuration.²⁰ This gives the intuitively reasonable result that the kinetic energy increases by twice the recoil energy $2E_r$ in each scattering event—this result can be derived directly from consideration of the conservation of energy and momentum in the scattering of photons (see Exercise 9.3).

Setting the time derivative equal to zero in eqn 9.24 gives the mean square velocity spread in the six-beam optical molasses configuration as

$$\overline{v_z^2} = 2E_r \frac{2R_{\text{scatt}}}{\alpha}, \quad (9.26)$$

and similarly along the other laser beam directions. The kinetic energy of the motion parallel to the z -axis is related to the temperature by $\frac{1}{2} M \overline{v_z^2} = \frac{1}{2} k_B T$ (according to the equipartition theorem). Substitution for α and R_{scatt} gives²¹

$$k_B T = \frac{\hbar \Gamma}{4} \frac{1 + (2\delta/\Gamma)^2}{-2\delta/\Gamma}. \quad (9.27)$$

¹⁸An underlying assumption here is that the absorption from the laser beams is uncorrelated, so that two laser beams produce twice as much diffusion as a single beam (given in eqn 9.23). This is only a reasonable approximation at low intensities ($I \ll I_{\text{sat}}$) where saturation is not significant, and, as in the derivation of eqn 9.17, we shall ignore the factor I/I_{sat} in the denominator of R_{scatt} . A more comprehensive treatment is given by Cohen-Tannoudji *et al.* (1992).

¹⁹Theoretically, this might seem difficult to achieve since, even if all three pairs of beams have the same polarization, the resultant electric field depends on the relative phase between the pairs of beams. In practice, however, these phases normally vary randomly in time (and with position if the beams are not perfectly aligned), so assuming that the light field is symmetrical over an average of many measurements is not too bad.

²⁰An alternative justification comes from considering the additional contribution along the z -axis from photons spontaneously emitted after absorption from the beams along the x - and y -axes. This contribution makes up for the fraction $1 - \eta$ of the spontaneous emission that goes in other directions following absorption from beams parallel to the z -axis. There is detailed balancing between different directions because of the symmetry of the configuration.

²¹Equation 9.17 can be written as

$$\begin{aligned} \alpha &= 2\hbar k^2 \frac{\partial R_{\text{scatt}}}{\partial \omega} \\ &= 2\hbar k^2 \frac{-2\delta}{\delta^2 + \Gamma^2/4} R_{\text{scatt}}. \end{aligned}$$

This function of $x = -2\delta/\Gamma$ has a minimum at $\delta = \omega - \omega_0 = -\Gamma/2$ of

$$k_B T_D = \frac{\hbar\Gamma}{2}. \quad (9.28)$$

This key result is the *Doppler cooling limit*. It gives the lowest temperature expected in the optical molasses technique. On general grounds, we expect a limit of this magnitude for processes in a two-level atom since $\hbar\Gamma = \hbar/\tau$ represents the smallest energy scale in the system.²² For sodium $T_D = 240\ \mu\text{K}$, which corresponds to a most probable velocity of $0.5\ \text{m s}^{-1}$. This velocity can be written as

$$v_D \simeq \left(\frac{\hbar\Gamma}{M}\right)^{1/2} = \left(\frac{\hbar k}{M} \cdot \frac{\Gamma}{k}\right)^{1/2} = (v_r v_c)^{1/2}, \quad (9.29)$$

where $v_c \simeq \Gamma/k$ gives an estimate (to within a factor of 2) of the capture velocity for the optical molasses technique, i.e. the velocity range over which F_{scatt} has a significant value. For sodium $v_r = 0.03\ \text{m s}^{-1}$ and $v_c = 6\ \text{m s}^{-1}$, and the above treatment of the optical molasses technique is valid for velocities within this range.²³

The theory as presented so far was initially thought to describe the optical molasses technique until experimental measurements found much lower temperatures under certain conditions, in particular when the Earth's magnetic field was cancelled out. The two-level model of an atom cannot explain this *sub-Doppler cooling*. Real alkali atoms have degenerate energy levels ($|JFM_F\rangle$ states). Remarkably, this does not just complicate the situation but actually allows new cooling mechanisms to occur, as described in Section 9.7. This is a rare example in which things turned out to be much better than expected. The fact that Doppler cooling theory does not accurately describe the optical molasses experiments with alkali metal atoms gives an excuse for the rather cavalier treatment of saturation in this section.

9.4 The magneto-optical trap

In the optical molasses technique cold atoms accumulate in the region where the three orthogonal pairs of laser beams intersect because it takes a considerable time for atoms to diffuse out, e.g. several seconds for beams of 1 cm radius. With the correct choice of polarizations for the laser beams, this configuration can be turned into a trap by the addition of a magnetic field gradient, as illustrated in Figs 9.8 and 9.9; the two coils with currents in opposite directions produce a quadrupole magnetic field. This magnetic field is much weaker than in the purely magnetic traps described in Chapter 10 and *does not* confine atoms by itself. In the *magneto-optical trap* (MOT) the quadrupole magnetic field causes an imbalance in the scattering forces of the laser beams and it is the radiation force that strongly confines the atoms.²⁴ The principle of the MOT is illustrated in Fig. 9.9(a) for a simple $J = 0$ to $J = 1$ transition. At the point in the middle of the coils the magnetic fields

²²Einstein pointed out in his fundamental work on radiative absorption and emission processes of atoms in a thermal radiation field that the momentum exchange between light and matter would bring the atoms into thermal equilibrium with the surroundings (Einstein 1917). When the radiation has a spectral distribution corresponding to 0 K (monochromatic light) the atom would be expected to approach this temperature.

²³Narrow transitions with $\hbar\Gamma < E_r$ give rise to a different behaviour, that has some similarities to the discussion of cooling using narrow Raman transitions in Section 9.8.

²⁴The principal idea of magneto-optical trapping was suggested by Jean Dalibard and demonstrated at Bell Laboratories, USA in collaboration with a group from MIT.

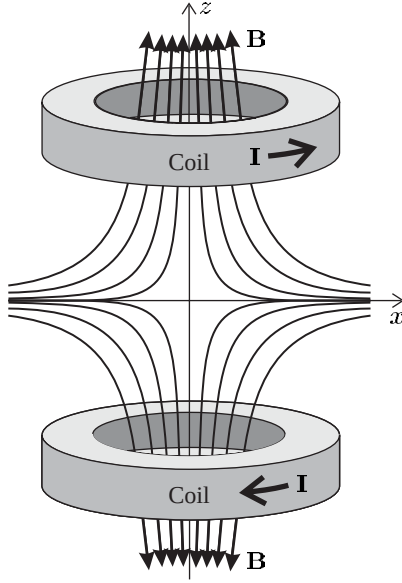


Fig. 9.8 A pair of coils with currents in opposite directions produces a quadrupole magnetic field. The field is zero at the centre of the coils and its magnitude increases linearly in every direction for small displacements from the zero point.

produced by the coils cancel out, so that $B = 0$. Close to this zero of the field there is a uniform field gradient that perturbs the atomic energy levels; the Zeeman effect causes the energy of the three sub-levels (with $M_J = 0, \pm 1$) of the $J = 1$ level to vary linearly with the atom's position, as shown for the z -axis in Fig. 9.9(a).²⁵ The counter-propagating laser beams have circular polarization as shown in Fig. 9.9(b) and a frequency slightly less than the atomic resonance frequency. The Zeeman shift causes an imbalance in the radiation force in the following way. Consider an atom displaced from the centre of the trap along the z -axis with $z > 0$, so the $\Delta M_J = -1$ transition moves closer to resonance with the laser frequency—the laser has a frequency below the atomic resonance in zero field to give damping by the optical molasses mechanism.²⁶ The selection rules lead to absorption of photons from the beam that excites the σ^- transition and this gives a scattering force that pushes the atom back towards the trap centre. A similar process occurs for a displacement in the opposite direction ($z < 0$); in this case the Zeeman shift of the transition frequency and selection rules favour absorption from the beam propagating in the positive z -direction that pushes the atom back towards $z = 0$. Note that these beam polarizations and the quantisation axis of the atom have been defined relative to a fixed direction in space, i.e. the z -direction in this one-dimensional example. For $z > 0$ this is the same as the direction of the magnetic field, but for $z < 0$ the magnetic field points the opposite way; hence the $M_J = -1$ state lies above $+1$ in this region, as shown in Fig. 9.9(a). Strictly speaking, σ^+ and σ^- refer to transitions of the atom and labelling the radiation as σ^+ is shorthand for circularly-polarized radiation of the handedness

²⁵The energy levels also vary in the other directions. The Maxwell equation $\text{div } \mathbf{B} = 0$ implies that

$$\frac{dB_x}{dx} = \frac{dB_y}{dy} = -\frac{1}{2} \frac{dB_z}{dz},$$

so the gradient in any radial direction is half of that along the z -direction.

²⁶The MOT requires three orthogonal pairs of $\sigma^+ - \sigma^-$ beams, but the optical molasses technique works with other polarization states, e.g. the Sisyphus cooling in Section 9.7 uses linearly-polarized beams.

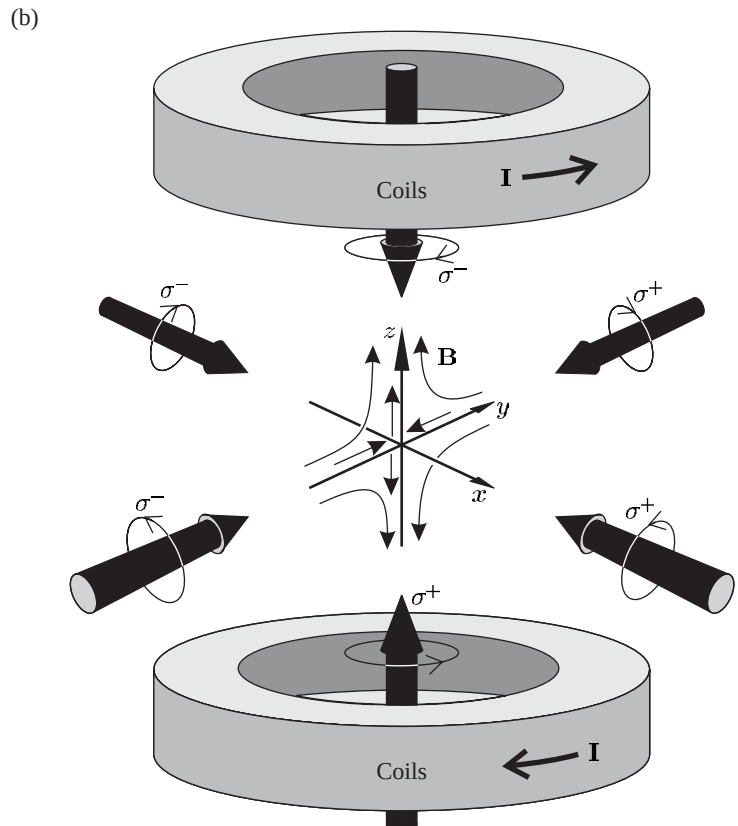
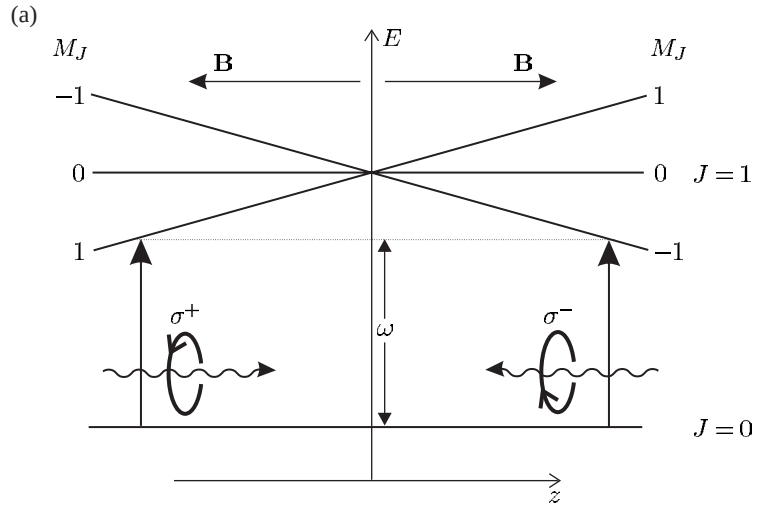


Fig. 9.9 (a) The mechanism of a magneto-optical trap illustrated for the case of an atom with a $J = 0$ to $J = 1$ transition. In the magnetic field gradient the Zeeman splitting of the sub-levels depends on the atom's position. Two counter-propagating beams of circularly-polarized light illuminate the atom and the selection rules for transitions between the Zeeman states lead to an imbalance in the radiative force from the laser beams that pushes the atom back towards the centre of the trap. (Not to scale; the Zeeman energy is much smaller than the optical transition energy.) (b) A magneto-optical trap is formed from three orthogonal pairs of laser beams, as in the optical molasses technique, that have the requisite circular polarization states and intersect at the centre of a pair of coils with opposite currents. The small arrows indicate the direction of the quadrupole magnetic field produced by the coils (as shown in more detail in Fig 9.8).

that excites the σ^+ transition (and similarly for σ^-).²⁷ To describe the magneto-optical trap mathematically we can incorporate the frequency shift caused by the Zeeman effect into eqn 9.15 (that describes the optical molasses technique):²⁸

$$F_{\text{MOT}} = F_{\text{scatt}}^{\sigma^+}(\omega - kv - (\omega_0 + \beta z)) - F_{\text{scatt}}^{\sigma^-}(\omega + kv - (\omega_0 - \beta z)) \\ \simeq -2\frac{\partial F}{\partial \omega}kv + 2\frac{\partial F}{\partial \omega_0}\beta z. \quad (9.30)$$

The term $\omega_0 + \beta z$ is the resonant absorption frequency for the $\Delta M_J = +1$ transition at position z , and $\omega_0 - \beta z$ is that for $\Delta M_J = -1$. The Zeeman shift at displacement z is

$$\beta z = \frac{g\mu_B}{\hbar} \frac{dB}{dz} z, \quad (9.31)$$

where $g = g_J$ in this case.²⁹ The force depends on the frequency detuning $\delta = \omega - \omega_0$, so $\partial F/\partial \omega_0 = -\partial F/\partial \omega$ and hence

$$F_{\text{MOT}} = -2\frac{\partial F}{\partial \omega}(kv + \beta z) \\ = -\alpha v - \frac{\alpha\beta}{k}z. \quad (9.32)$$

The imbalance in the radiation force caused by the Zeeman effect leads to a restoring force with spring constant $\alpha\beta/k$ (which is written in this form to emphasise that it arises in a similar way to damping). Under typical operating conditions the atom undergoes over-damped simple harmonic motion, as shown in Exercise 9.9. Atoms that enter the region of intersection of the laser beams are slowed (as in the optical molasses technique) and the position-dependent force pushes the cold atoms to the trap centre. This combination of strong damping and trapping makes the magneto-optical trap easy to load and it is very widely used in laser cooling experiments.

A typical apparatus uses an MOT to collect cold atoms from a slowed atomic beam. When sufficient atoms have accumulated the magnetic field of the MOT is turned off to cool the atoms by the optical molasses technique before further experiments are carried out.³⁰ This procedure gives more atoms (at a higher density) than the optical molasses technique on its own because the MOT captures faster atoms than optical molasses. The magnetic field in the MOT changes the atom's absorption frequency in a similar way to the Zeeman slowing technique, e.g. if the magneto-optical trap has laser beams of radius 5 mm and we take this as the stopping distance in eqn 9.8 then the trap captures sodium atoms with velocities less than $v_c(\text{MOT}) \simeq 70 \text{ m s}^{-1}$. But atoms enter the MOT from all directions and the magnetic field varies linearly with position (a constant gradient), so the situation is not the same as the optimum case of an atom moving along the axis of a tapered solenoid with the counter-propagating laser beam (see Exercise 9.10). Nevertheless, an MOT captures atoms with much faster velocities than the optical

²⁷This is a convenient convention for discussing the principles of laser cooling where the transitions that occur depend on the sense of rotation of the electric field around the quantization axis of the atom (whereas handedness depends on both the sense of the rotation and the direction of the propagation). The electric field of the circularly-polarized radiation drives the bound atomic electron(s) around in the same sense as the electric field; therefore, radiation labelled σ^+ that imparts positive angular momentum about the quantization axis, $\Delta M_J = +1$, has an electric field that rotates clockwise when viewed along that quantization axis (see Fig. 9.9(b)), i.e. in the same direction as a particle with $\langle L_z \rangle > 0$. The magneto-optical trap also confines atoms along the x - and y -axes, and all other directions. In practice, these traps are extremely robust and only require the polarizations of the beams to be approximately correct—some atoms are trapped so long as none of the beams have the wrong handedness.

²⁸This assumes a small Zeeman shift $\beta z \ll \Gamma$ in addition to the small velocity approximation $kv \ll \Gamma$.

²⁹More generally, $g = g_{F'}M_{F'} - g_F M_F$ for a transition between the hyperfine-structure levels $|F, M_F\rangle$ and $|F', M_{F'}\rangle$; however, $g \simeq 1$ for many of the transitions used for laser cooling—see Example 9.1.

³⁰Atoms in the MOT have a higher temperature than in the optical molasses technique for several reasons: the sub-Doppler cooling mechanisms break down when the Zeeman shift exceeds the light shift and there is strong absorption of the laser beams as they pass through dense clouds of cold atoms (see Exercise 9.12).

³¹These elements have an appreciable vapour pressure at room temperature. The MOT captures the slowest atoms in the Maxwellian velocity distribution. The equilibrium number of atoms trapped directly into an MOT from a vapour is proportional to the fourth power of $v_c(\text{MOT})$, so the method is very sensitive to this parameter.

³²Broadening from collisions and the Doppler effect is negligible and broadening caused by the inhomogeneous magnetic field is small, e.g. for a typical field gradient of 0.1 T m^{-1} and a cloud of radius 3 mm the variation in the Zeeman shift is $\sim 4 \text{ MHz}$ (for $g = 1$).

molasses technique, e.g. for sodium $v_c(\text{MOT}) > v_c(\text{molasses}) \simeq 6 \text{ m s}^{-1}$ (eqn 9.29). This relatively large capture velocity makes it possible to load an MOT directly from a room temperature vapour, and this method can be used instead of slowing an atomic beam for the heavy alkalis rubidium and caesium (see Exercise 9.11).³¹ Typically, an MOT loaded from a slow atomic beam contains up to 10^{10} atoms. Experiments that capture atoms directly from a vapour usually have considerably less. Such general statements should be treated with caution, however, since there are various factors that limit number and density in different operating regimes, e.g. absorption of the laser light—the cold atoms that congregate in the centre of the MOT have close-to-the-maximum optical absorption cross-section³²—absorption leads to a difference or an imbalance in the intensities of the laser beams propagating through the cloud of cold atoms that affects the trapping and cooling mechanisms.

At equilibrium each atom absorbs and emits the same amount of light. Therefore a large cloud of cold atoms in an MOT scatters a significant fraction of the incident light and the atoms can be seen with the naked eye as a bright glowing ball in the case of sodium; for rubidium the scattered infra-red radiation can easily be detected on a CCD camera. The MOT provides a source of cold atoms for a variety of experiments, e.g. loading the dipole-force traps (as described in the following sections) and magnetic traps (Chapter 10). Finally, it is worth highlighting the difference between magneto-optical and magnetic trapping. The force in the MOT comes from the radiation—the atoms experience a force close to the maximum value of the scattering force at large displacements from the centre. The magnetic field gradients in a magneto-optical trap (that tune the absorption frequency of the atoms) are much smaller than those used in magnetic traps. A typical MOT has a gradient of 0.1 T m^{-1} and when the light is switched off this produces a magnetic force that is not sufficient to support atoms against gravity.

9.5 Introduction to the dipole force

The scattering force equals the rate at which an object gains momentum as it absorbs radiation. Another type of radiation force arises from the refraction of light as illustrated in Fig. 9.10. A simple prism that deflects light through an angle θ feels a force

$$F = \left(\frac{IA}{c} \right) 2 \sin \left(\frac{\theta}{2} \right), \quad (9.33)$$

where the quantity IA/c corresponds to the rate at which radiation with intensity I carries momentum through a cross-sectional area A (perpendicular to the direction of propagation); this quantity corresponds to the total force when the radiation is absorbed (eqn 9.1). When the beam is refracted the difference between the incoming and outgoing momentum flow leads to the factor $2 \sin(\theta/2)$ by simple geometry. The angle and the resultant force increase with the refractive index.

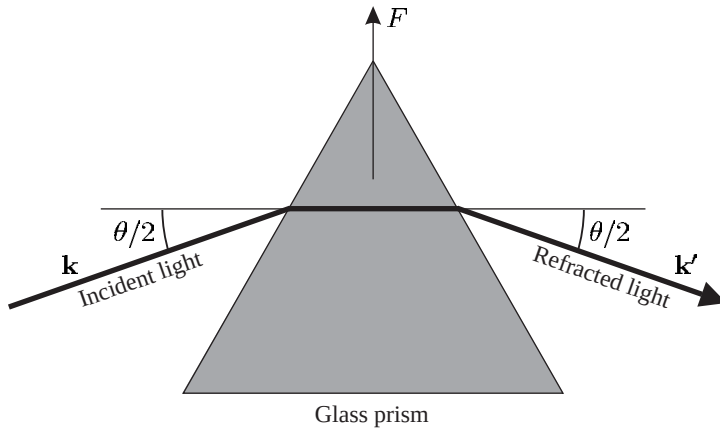


Fig. 9.10 Radiation that is deflected by a glass prism (or a mirror) exerts a force on that object equal and opposite to the rate of change of momentum of the radiation.

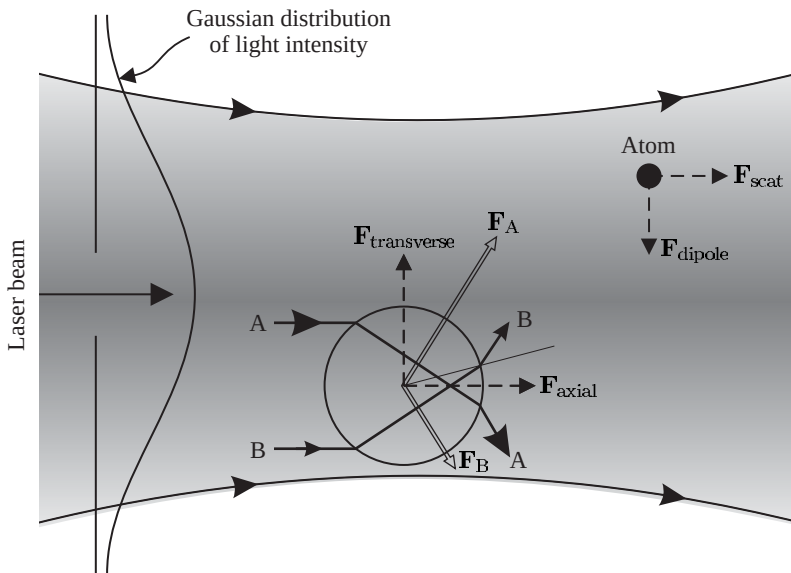


Fig. 9.11 The refraction of light by a dielectric sphere with a refractive index greater than that of the surrounding medium. In a laser beam with a Gaussian profile the intensity along ray A is greater than for ray B. This leads to a resultant force on the sphere towards the region of high intensity (centre of the laser beam), in addition to an axial force pushing in the direction of the beam. Analogous forces arise on smaller particles such as atoms. After Ashkin (1997).

These simple considerations show that the forces associated with absorption and refraction by an object have similar magnitude but they have different characteristics; this can be seen by considering a small dielectric sphere that acts as a converging lens with a short focal length, as shown in Fig. 9.11. In a laser beam of non-uniform intensity, the difference in the intensity of the light refracted on opposite sides of the sphere leads to a resultant force that depends on the gradient of the intensity: a sphere with a refractive index greater than the surrounding medium, $n_{\text{sphere}} > n_{\text{medium}}$, feels a force in the direction of increasing intensity, whereas a sphere with $n_{\text{sphere}} < n_{\text{medium}}$ is pushed away from the region of high intensity.³³ Thus the sign of this *gradient force* (also known as the dipole force) depends on n_{sphere} . The refractive index of materials varies with frequency in the characteristic way shown in Fig. 9.12. This behaviour can be understood in terms of a simple classi-

³³The calculation of this force using geometrical optics is straightforward in principle, but integration over all the different rays and inclusion of reflection coefficients makes it complicated.

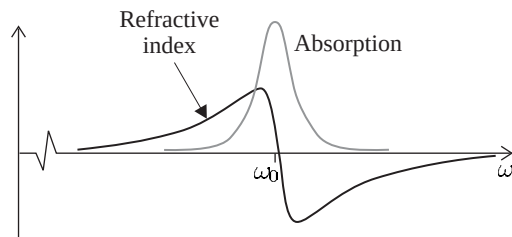


Fig. 9.12 Absorption has a Lorentzian line shape with a peak at the resonance frequency ω_0 . The refractive index is zero on resonance, where it changes sign, and this characteristic dependence on frequency leads to dispersion.

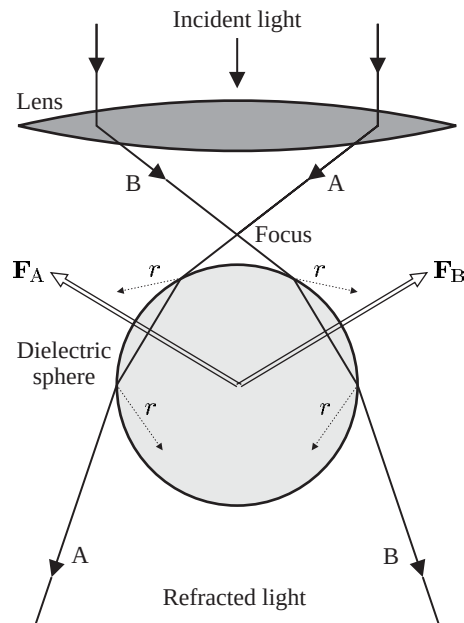


Fig. 9.13 A tightly-focused beam of light exerts a radiation force on a dielectric sphere that pulls it towards the region of high intensity (at the focus). Not all of the light is transmitted at the interfaces and the reflected rays are indicated. After Ashkin (1997).

cal model in which bound electrons execute damped harmonic oscillation with resonance frequency ω_0 , see Section 7.5.1 and Fox (2001), but the relationship between the refractive index and absorption at the resonant frequency is very general (independent of any particular model). Dispersion and absorption are different facets of the same interaction of light with matter; strong absorption leads to large changes in refractive index. The variation in the refractive index extends over a larger frequency range than the absorption, e.g. although air and glass (of good optical quality) are both transparent at visible wavelengths they have refractive indices of 1.0003 and 1.5, respectively, associated with strong absorption in the ultraviolet region; $n_{\text{glass}} - 1 \gg n_{\text{air}} - 1$ because a solid has a higher density of atoms than a gas.³⁴

The force that attracts an object towards a region of high intensity has been used to manipulate microscopic objects in a technique called optical tweezers that was developed by Arthur Ashkin (Ashkin *et al.* 1986). The objective lens in an optical microscope is used to focus a laser beam tightly so that there is a strong gradient force along the axis, as shown in Fig. 9.13, in addition to trapping in the radial direction

³⁴Generally speaking, the effects of refraction are most apparent when they are not obscured by absorption, i.e. away from a resonance. A similar situation arises for forces on the individual atoms.

shown in Fig. 9.11. As the laser beam is moved the particles remain trapped in the region of high intensity. The microscope is used to view the object through a filter that blocks the laser light. The objects are immersed in water and mounted on a microscope slide in a standard way. The liquid provides viscous damping of the motion.³⁵ Optical tweezers works, not just for simple spheres, but also for biological cells such as bacteria, and these living objects can withstand the focused intensity required to trap them without harm (the surrounding water prevents the cells from heating up). For example, experiments have been carried out where a bacterium is tethered to the surface of a glass microscope slide by its flagellum (or ‘tail’) and the body is moved around by optical tweezers. This gives a quantitative measure of the force produced by the microscopic biological motor that moves the flagellum to propel these organisms (see Ashkin (1997) and Lang and Bloch (2003) for reviews).

This section has introduced the concept of a radiation force called the gradient force, or dipole force, and the next section shows that a similar force occurs for atoms.³⁶

³⁵Laser radiation can levitate small objects in air, but this is less straightforward than the manipulation of objects floating in water.

³⁶This analogy is not just of pedagogical interest—the first experiments on optical tweezers and the dipole-force trapping of atoms were carried out in the same place (Bell Laboratories in the USA).

9.6 Theory of the dipole force

Actually, this section does not just derive the dipole force on an atom from first principles but also the scattering force, and so demonstrates the relationship between these two types of radiation force. An electric field $\mathbf{E}$ induces a dipole moment of $-\mathbf{er} = \epsilon_0\chi_a\mathbf{E}$ on an atom with a (scalar) polarizability $\epsilon_0\chi_a$. The interaction energy of this dipole with the electric field is given by

$$U = -\frac{1}{2}\epsilon_0\chi_a E^2 = \frac{1}{2}\mathbf{er} \cdot \mathbf{E}, \quad (9.34)$$

where E is the amplitude of the electric field and U is used here to denote energy to avoid confusion with the electric field. This expression comes from the integration of $dU = -\epsilon_0\chi_a\mathcal{E} d\mathcal{E}$ from $\mathcal{E} = 0$ to $\mathcal{E} = E$ (the factor of 1/2 does not occur for a permanent electric dipole). Differentiation gives the z -component of the force as

$$F_z = -\frac{\partial U}{\partial z} = \epsilon_0\chi_a E \frac{\partial E}{\partial z}, \quad (9.35)$$

and similarly for F_x and F_y . Radiation of angular frequency ω , propagating along the z -direction, can be modelled as an electric field $\mathbf{E} = E_0 \cos(\omega t - kz) \hat{\mathbf{e}}_x$.³⁷ The gradient of the energy gives the z -component of the force as³⁸

$$F_z = -ex \left\{ \frac{\partial E_0}{\partial z} \cos(\omega t - kz) + kE_0 \sin(\omega t - kz) \right\}. \quad (9.36)$$

The two parts of this force can be understood using either the classical or the quantum mechanical expressions for the dipole that were derived in Chapter 7. The classical model of the atom as an electron undergoing

³⁷This particular field is linearly polarized parallel to $\hat{\mathbf{e}}_x$, as in Section 7.3.2, but the results derived here are quite general.

³⁸This classical treatment gives the same result as the quantum mechanical derivation when the electric field varies slowly over the typical dimensions of an atomic wavepacket ($\lambda_{dB} \ll \lambda_{light}$). Under these circumstances, classical equations of motion correspond to equations for the expectation values of the quantum operators, e.g. the rate of change of momentum equals the force corresponding to

$$\frac{d\langle \mathbf{p} \rangle}{dt} = -\langle \nabla U \rangle.$$

This is an example of Ehrenfest’s theorem in quantum mechanics. The quantum mechanical derivation of the dipole force is given in Cohen-Tannoudji *et al.* (1992).

³⁹As shown in Section 7.5.1, the classical model does not account for saturation.

simple harmonic motion gives very useful insight into the frequency dependence of the force, but the quantum treatment is required to find the correct intensity dependence.³⁹ This section presents both approaches, starting with the classical one—because of the very close parallels between them this requires little extra effort.

Classically, the displacement of the electron x by an electric field is calculated by modelling the atom as a harmonic oscillator with a driving term. Expressing x in terms of its components in phase ($\mathcal{U}$) and in quadrature ($\mathcal{V}$) to the applied field (cf. eqn 7.56), we find

$$F_z = -e \{ \mathcal{U} \cos(\omega t - kz) - \mathcal{V} \sin(\omega t - kz) \} \times \left\{ \frac{\partial E_0}{\partial z} \cos(\omega t - kz) + E_0 k \sin(\omega t - kz) \right\}. \quad (9.37)$$

⁴⁰Using $\overline{\sin^2} = \overline{\cos^2} = \frac{1}{2}$.

The time average over many oscillation periods gives⁴⁰

$$\begin{aligned} \overline{F_z} &= \frac{-e}{2} \left\{ \mathcal{U} \frac{\partial E_0}{\partial z} - \mathcal{V} k E_0 \right\} \\ &= \frac{e^2}{4m\omega} \left\{ \frac{-(\omega - \omega_0) E_0}{(\omega - \omega_0)^2 + (\beta/2)^2} \frac{\partial E_0}{\partial z} + \frac{(\beta/2) k E_0^2}{(\omega - \omega_0)^2 + (\beta/2)^2} \right\}, \end{aligned} \quad (9.38)$$

using eqns 7.58 and 7.59 for $\mathcal{U}$ and $\mathcal{V}$. The intensity of the light is $I = \frac{1}{2} \epsilon_0 c E_0^2$ and, by a simple extension of the derivation given above to the x - and y -directions, the radiation force can be written in vector notation as

$$\overline{\mathbf{F}} = \frac{e^2}{2\epsilon_0 m c} \left\{ \frac{-(\omega - \omega_0)}{(\omega - \omega_0)^2 + (\beta/2)^2} \frac{\nabla I}{\omega} + \frac{\beta/2}{(\omega - \omega_0)^2 + (\beta/2)^2} \frac{I}{c} \frac{\mathbf{k}}{|\mathbf{k}|} \right\}. \quad (9.39)$$

The in-phase component of the dipole ($\mathcal{U}$) leads to a force proportional to the gradient of the intensity. The frequency dependence of this component follows a dispersive line shape that is closely related to the refractive index,⁴¹ as shown in Fig. 9.12. (The dependence on $1/\omega$ has a negligible effect on narrow transitions $\beta \ll \omega_0$.) At the atomic resonance frequency $\omega = \omega_0$ the component $\mathcal{U} = 0$. The quadrature term, from $\mathcal{V}$, has a Lorentzian line shape and this force, arising from absorption, is proportional to I and points along the wavevector of the radiation $\mathbf{k}$. This classical model gives a simple way of understanding various important features of the forces on atoms and shows how they relate to radiation forces on larger objects (such as those discussed in the introductory Sections 9.1 and 9.5); however, we shall not use it for quantitative calculations.

To find the force quantum mechanically, we use eqn 7.36 for the dipole moment in terms of the components of the Bloch vector u and v . Substitution into eqn 9.36, and taking the time average as above, gives (cf. eqn 9.38)

$$\overline{F_z} = \frac{-eX_{12}}{2} \left\{ u \frac{\partial E_0}{\partial z} - v E_0 k \right\} \quad (9.40)$$

$$= F_{\text{dipole}} + F_{\text{scatt}}. \quad (9.41)$$

⁴¹In optics, it is generally the effect of the medium on the light that is of interest, e.g. the angle through which the medium refracts, or bends, a light beam, but this implies that the medium feels a force equal to the rate of change of the momentum of the light. The refractive index, and absorption coefficient, describe bulk properties, whereas it is the effect of light on individual atoms that is of interest here.

The force that depends on the in-phase component of the dipole u is the dipole force and the other part is the scattering force. Using the expressions for u and v given in eqn 7.68 and the Rabi frequency $\Omega = eX_{12}E_0/\hbar$, we find that

$$F_{\text{scatt}} = \hbar k \frac{\Gamma}{2} \frac{\Omega^2/2}{\delta^2 + \Omega^2/2 + \Gamma^2/4}, \quad (9.42)$$

which is consistent with eqn 9.4, and

$$F_{\text{dipole}} = -\frac{\hbar\delta}{2} \frac{\Omega}{\delta^2 + \Omega^2/2 + \Gamma^2/4} \frac{\partial\Omega}{\partial z}, \quad (9.43)$$

where $\delta = \omega - \omega_0$ is the frequency detuning from resonance. The expression for the scattering force has been repeated here for ease of comparison with eqn 9.43. These forces have essentially the same frequency dependence as in the classical model, with a line width that is power broadened so that $\beta \longleftrightarrow \Gamma(1 + 2\Omega^2/\Gamma^2)^{1/2}$. The dipole force is zero on resonance ($F_{\text{dipole}} = 0$ for $\delta = 0$), and for $|\delta| \gg \Gamma$ (and an intensity such that $|\delta| \gg \Omega$) the dipole force equals the derivative of the light shift (eqn 7.93):

$$F_{\text{dipole}} \simeq -\frac{\partial}{\partial z} \left(\frac{\hbar\Omega^2}{4\delta} \right). \quad (9.44)$$

Thus the light shift, or a.c. Stark shift, for an atom in the ground state acts as a potential U_{dipole} in which the atom moves. More generally, in three dimensions

$$\mathbf{F}_{\text{dipole}} = -\left\{ \hat{\mathbf{e}}_x \frac{\partial}{\partial x} + \hat{\mathbf{e}}_y \frac{\partial}{\partial y} + \hat{\mathbf{e}}_z \frac{\partial}{\partial z} \right\} U_{\text{dipole}} = -\nabla U_{\text{dipole}}, \quad (9.45)$$

where

$$U_{\text{dipole}} \simeq \frac{\hbar\Omega^2}{4\delta} \equiv \frac{\hbar\Gamma}{8} \frac{\Gamma}{\delta} \frac{I}{I_{\text{sat}}}. \quad (9.46)$$

When δ is positive ($\omega > \omega_0$) this potential has a maximum where the intensity is highest—the atom is repelled from regions of high intensity. In the opposite case of frequency detuning to the red (δ negative) the dipole force acts in the direction of increasing I , and U_{dipole} is an attractive potential—atoms in a tightly-focused laser beam are attracted towards the region of high intensity, both in the radial direction and along the axis of the beam. This dipole force confines atoms at the focus of a laser beam in an analogous way to optical tweezers to create a **dipole-force trap**.⁴² Normally, dipole traps operate at large frequency detuning ($|\delta| \gg \Gamma$), where to a good approximation eqn 9.3 becomes

$$R_{\text{scatt}} \simeq \frac{\Gamma}{8} \frac{\Gamma^2}{\delta^2} \frac{I}{I_{\text{sat}}}. \quad (9.47)$$

This scattering rate depends on I/δ^2 , whereas the trap depth is proportional to I/δ (in eqn 9.46). Thus working at a sufficiently large frequency detuning reduces the scattering whilst maintaining a reasonable trap depth (for a high intensity at the focus of the laser beam). Usually

⁴²The situation for an atom with detuning $\delta < 0$ resembles that of a dielectric sphere with a refractive index greater than the surrounding medium.

there are two important criteria in the design of dipole-force traps: (a) the trap must be deep enough to confine the atoms at a certain temperature (that depends on the method of cooling); and (b) the scattering rate must be low to reduce heating.

Example 9.2 *Dipole trapping of sodium atoms*

The wavelength of the laser light used for a dipole-force trap depends mainly on practical considerations.⁴³ It is convenient to use a high-power solid-state laser such as a Neodymium:YAG laser that produces continuous-wave radiation at a fixed infra-red wavelength of $\lambda = 1.06 \mu\text{m}$. The frequency detuning of this laser radiation from the sodium resonance at $\lambda_0 = 589 \text{ nm}$ is

$$\frac{\delta}{\Gamma} = \frac{2\pi}{\Gamma} \left\{ \frac{c}{\lambda_0} - \frac{c}{\lambda} \right\} = 2.3 \times 10^7, \quad (9.48)$$

⁴³Whereas for scattering-force techniques the frequency of the laser must be tunable, so that it can be adjusted to within several line widths from an atomic transition frequency.

in units of Γ , where $1/\Gamma = \tau = 16 \text{ ns}$.⁴⁴ Solid-state lasers can produce powers of many tens of watts, but in this example we use a conservative value of $P = 1 \text{ W}$. When focused to a waist of $w_0 = 10 \mu\text{m}$ this laser beam has an intensity of $I = 2P/(\pi w_0^2) = 6.4 \times 10^9 \text{ W m}^{-2} \equiv 1 \times 10^8 I_{\text{sat}}$. Equation 9.46 gives

$$U_{\text{dipole}} = \frac{\hbar\Gamma}{2} \times 1.1 = 260 \mu\text{K}. \quad (9.49)$$

Thus atoms cooled below the Doppler cooling limit $\hbar\Gamma/2$ can be trapped. For this laser intensity and frequency detuning, eqn 9.47 gives

$$R_{\text{scatt}} = 2.4 \times 10^{-8} \Gamma = 2 \text{ s}^{-1}. \quad (9.50)$$

⁴⁵If atoms gain twice the recoil energy per scattering event, as in eqn 9.24, it takes many seconds before the atoms boil out of the trap. The fluctuations in the dipole force itself can cause heating and $\mathbf{F}_{\text{dipole}} + \delta\mathbf{F}_{\text{dipole}}$ should be included in eqn 9.20. The fluctuations $\delta\mathbf{F}_{\text{dipole}}$ give comparatively small heating in a dipole-force trap with a large frequency detuning; however, there are circumstances where $\delta\mathbf{F}_{\text{dipole}}$ is the dominant cause of heating, e.g. for the Sisyphe effect described in Section 9.7.

⁴⁶Writing eqn 9.50 as

$$R_{\text{scatt}} = 5 \times 10^{-8} (\Gamma/2),$$

where $\Gamma/2$ is the maximum of R_{scatt} , shows that F_{scatt} is 5×10^{-8} times its maximum value.

A sodium atom only scatters a few photons per second which gives a low heating rate.⁴⁵ The scattering force is negligible for these conditions,⁴⁶ i.e. the force pushing in the direction of the light is weaker than the dipole force pulling the atom towards the high-intensity focus. The condition that the laser light has a frequency detuning far from the atomic resonance is not restrictive, and calculations along the same lines as that given here for sodium show that a laser with the above properties can be used for the dipole-force trapping of any alkali metal atom.

A force derived from a potential is conservative, i.e. the total energy remains constant during motion. Thus an atom that enters a dipole trap gains kinetic energy as it moves towards the bottom of the potential well and then it rides up the other side of the trap and escapes, because no energy is lost. To load a dipole trap there must be either some dissipation of energy by spontaneous emission (as in the MOT), or the atoms must be placed gently in the bottom of the trap. In the first experimental demonstration of a dipole trap for atoms the laser beam was focused into a cloud of atoms that were cooled by the optical molasses technique, see Fig. 9.14 (Chu *et al.* 1986). The trapped atoms were observed as a bright spot in the region of more diffuse fluorescence from

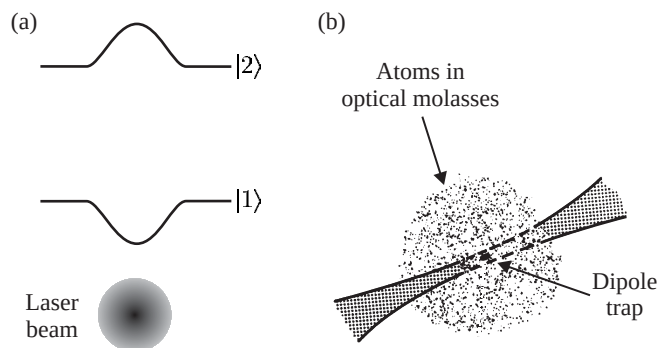


Fig. 9.14 (a) An intense laser beam alters the energy levels of an atom, as illustrated for a radial direction across a laser beam propagating perpendicular to the plane of the figure. For a laser frequency less than the atomic resonance frequency the a.c. Stark effect forms a potential well in the ground-state energy and atoms are attracted towards regions of high intensity. (b) The dipole-force trap formed by a focused laser beam can be loaded with cold atoms produced by the optical molasses technique, as described in the text.

the region of optical molasses because the density of trapped atoms was greater. When the focused laser beam was first switched on the dipole trap contained relatively few atoms within its small volume, at a density comparable with that in the surrounding region, e.g. 10^{10} cm^{-3} . This was perceived to be a problem, but atoms that started off outside the trap executed a random walk⁴⁷ that took some of them into the dipole trap, where they remained. In this way atoms accumulated in the trap to give a high density.

A dipole-force trap formed by a single laser beam gives tight radial confinement, but it is weak in the axial direction. Therefore the atoms in such a trap form an elongated, cigar-shaped cloud. To obtain strong confinement in all directions, if necessary, one can form a dipole-force trap at the intersection of two laser beams.⁴⁸ Many other configurations are possible and the design of dipole traps is restricted only by the form of the intensity distributions that can be sculpted from laser light. Diffraction limits the minimum distance over which the intensity of the light changes. An ingenious way of creating a high-intensity gradient is shown in Fig. 9.15. A laser beam that is totally internally reflected at the surface of glass gives an evanescent wave in which the electric field falls off exponentially over a distance of the wavelength of the light.⁴⁹ For a laser frequency to the blue ($\delta > 0$), the repulsive dipole force near the surface acts like a reflective coating for atoms. This creates a mirror that reflects low-energy atoms, as shown in Fig. 9.15.

9.6.1 Optical lattice

The dipole force is strong in a standing wave of light because the intensity changes from a maximum (at the anti-nodes) to zero (at the nodes) over a distance of $\lambda/2$ to give a high gradient of intensity. The physical explanation for this strong force is stimulated scattering of radiation. In a standing wave, an atom absorbs light with wavevector $\mathbf{k}$ from one beam and the laser beam in the opposite direction stimulates emission with wavevector $\mathbf{k}' = -\mathbf{k}$; this gives the atom an impulse of $2\mathbf{k}$. The rate of this stimulated process does not saturate at high intensities.⁵⁰

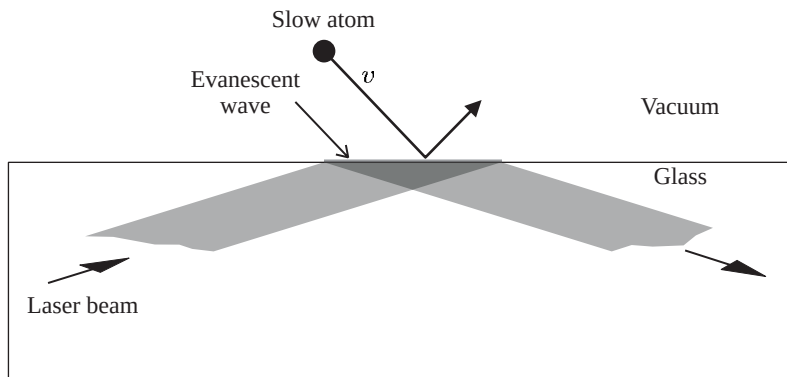
⁴⁷A random walk in space leading to spatial diffusion, rather than the random walk of the velocity leading to heating that was used to calculate the Doppler cooling limit—both processes are caused by scattering.

⁴⁸The dipole potential is proportional to the total intensity. Laser beams with orthogonal polarizations, or substantially different frequencies, do not interfere and the total intensity is the sum of the individual intensities.

⁴⁹This behaviour of the light closely resembles the quantum reflection at a potential step that is higher than the energy of the incident particle. The wavefunction falls exponentially to zero in the classically forbidden region.

⁵⁰More generally, the dipole force arises from a *stimulated* process of absorption of a photon of wavevector $\mathbf{k}_1$ and stimulated emission with wavevector $\mathbf{k}_2$. In this process the atom receives an impulse $\hbar(\mathbf{k}_1 - \mathbf{k}_2)$ that changes the atom's momentum. A tightly-focused laser beam contains a range of wavevectors and exerts a dipole force on an atom analogous to that in the optical tweezers technique. A dipole force cannot occur in a plane wave since the stimulated processes have $\mathbf{k}_1 = \mathbf{k}_2$.

Fig. 9.15 The evanescent wave, created when a laser beam is totally internally reflected, forms a mirror for atoms. For a light with a blue detuning ($\omega > \omega_0$) the dipole force repels atoms from the region of high intensity close to the vacuum–glass interface. (The same principle applies if the surface is curved (e.g. concave) and such an arrangement can be used to focus the matter waves.)



The dipole potential associated with this force depends on the intensity of the light (cf. eqn 9.46 for a large frequency detuning). Two counter-propagating beams of linearly-polarized light produce an electric field given by

$$\begin{aligned}\mathbf{E} &= E_0 \{ \cos(\omega t - kz) + \cos(\omega t + kz) \} \hat{\mathbf{e}}_x \\ &= 2E_0 \cos(kz) \cos(\omega t) \hat{\mathbf{e}}_x.\end{aligned}\quad (9.51)$$

⁵¹This form is only true for large detunings, $\delta \gg \Gamma, \Omega$. If this inequality is satisfied then there is also little spontaneous scattering from the atoms.

This standing wave gives a dipole potential of the form⁵¹

$$U_{\text{dipole}} = U_0 \cos^2(kz). \quad (9.52)$$

⁵²Interference will generally lead to a periodic arrangement of positions where atoms become localised (at intensity maxima for a frequency detuning to the red) in a three-dimensional standing wave of light, but the creation of a particular configuration of the optical lattice requires control of the polarization (and relative phase).

⁵³The phase-space density at which BEC occurs is approximately equal to that at which there is one atom per well in the ground state of an optical lattice.

⁵⁴As atoms pass through a standing wave of light they accumulate a phase shift of order $\phi \simeq U_0 t / \hbar$ for an interaction time t . Light with either sign of frequency detuning can be used to give $\phi \sim \pm\pi$, and so create a phase grating with significant amplitude in the diffraction orders.

Here U_0 is the light shift at the anti-nodes—these maxima have an intensity four times that of the individual beams. For a frequency detuning to the red, a standing wave of light traps atoms at the anti-nodes and gives confinement in the radial direction as in a single beam. This regular array of microscopic dipole traps is called an *optical lattice*. With more laser beams the interference between them can create a regular array of potential wells in three dimensions, e.g. the same configuration of six beams in the optical molasses technique shown in Fig. 9.5 (along $\pm\hat{\mathbf{e}}_x$, $\pm\hat{\mathbf{e}}_y$ and $\pm\hat{\mathbf{e}}_z$) can create a regular cubic lattice of potential wells for suitable polarizations and a large frequency detuning.⁵² The potential wells in this optical lattice have a spacing of $\lambda/2$, and so one atom per lattice site corresponds to a density of $8/\lambda^3 \simeq 7 \times 10^{13} \text{ cm}^{-3}$ for $\lambda = 1.06 \mu\text{m}$. Therefore the sites will be sparsely populated when the atoms are loaded into the lattice after cooling by the optical molasses technique. (A typical number density in the optical molasses technique is a few times $10^{10} \text{ cm}^{-3} \equiv 0.01 \text{ atoms } \mu\text{m}^{-3}$.)

Experiments that load more than one atom in each potential well have been carried out by adiabatically turning on the light that creates an optical lattice in a region containing a sample of atoms that are in a Bose–Einstein condensate (see Chapter 10).⁵³ Moreover, these atoms go into the lowest vibrational level in each of the potential wells. The use of one-dimensional standing waves as diffraction gratings for matter waves is discussed in Chapter 11.⁵⁴

9.7 The Sisyphus cooling technique

9.7.1 General remarks

The dipole force experienced by atoms in a light field can be stronger than the maximum scattering force because F_{dipole} does not saturate with increasing intensities (whereas F_{scatt} does), but stimulated processes alone cannot cool atoms. To dissipate energy there must be some spontaneous emission to carry away energy from the atoms—this is true for all cooling mechanisms, e.g. Doppler cooling by the scattering force, and it is particularly apparent for the process described in this section.

The first experimental evidence that Doppler cooling does not give a complete description of the laser cooling in a standing wave came from measurements of the velocity distribution of atoms by the direct time-of-flight method shown in Fig. 9.16. When William Phillips and co-workers carried out such measurements they were pleasantly surprised to find that the optical molasses technique can cool atoms below the Doppler

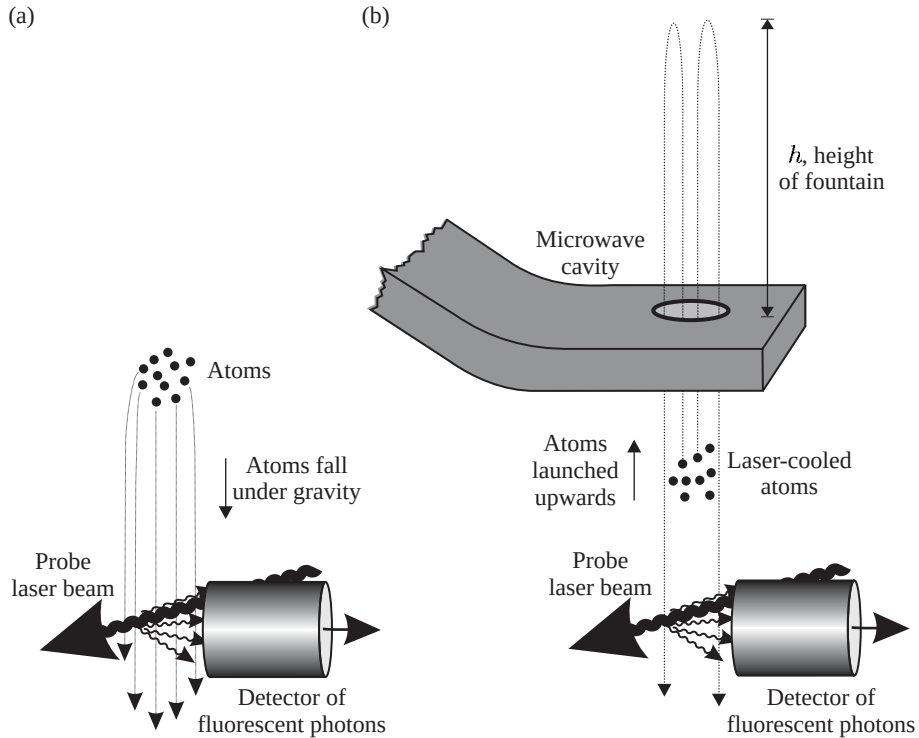


Fig. 9.16 (a) The temperature of a sample of atoms that has been cooled by the optical molasses technique is measured by turning off the six laser beams (not shown) so that the cloud of cold atoms falls downwards to the bottom of the vacuum chamber (because of gravity). The expansion of the falling cloud depends on the initial spread of the velocities. To observe this expansion a horizontal probe laser beam is aligned several centimetres below the initial position of the cloud. This probe beam has a frequency close to the atomic resonance frequency $\omega \simeq \omega_0$ so that atoms scatter light as they pass through and this fluorescence signal is recorded (or absorption could be monitored). (b) Instead of just dropping the atoms, they can be launched upwards to form an atomic fountain. This configuration is used for precision measurements, as described in Section 9.9.

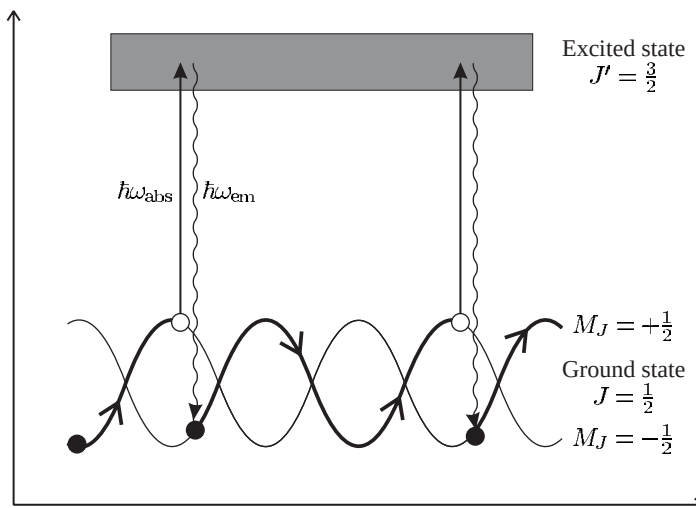
cooling limit in eqn 9.28. This *cannot* be understood in terms of a simple picture in which the scattering forces from each of the six laser beams add independently. Sodium, and other alkalis, have Zeeman structure in their ground states and this additional complexity, as compared to a simple two-level atom, allows new processes to occur.

A particularly important mechanism by which atoms dissipate energy as they move through a standing wave is the **Sisyphus effect** that was explained by Jean Dalibard and Claude Cohen-Tannoudji (1989), and this section follows the description given in that seminal paper. Steven Chu and co-workers also developed a model to explain sub-Doppler cooling based on the transfer of population between the different sub-levels of the ground configuration (optical pumping) as atoms move through the light field. This transfer of populations takes place on a time-scale τ_{pump} that can be much longer than the spontaneous lifetime ($\tau_{\text{pump}} \gg 1/\Gamma$). This longer time-scale gives better energy resolution than in a two-level atom and therefore allows cooling below the Doppler cooling limit, i.e. $k_B T \simeq \hbar/\tau_{\text{pump}} < \hbar\Gamma/2$. A specific example of this general argument is shown in Fig. 9.17, and the following section gives more details.

9.7.2 Detailed description of Sisyphus cooling

Consider an atom that has a lower level with angular momentum $J = 1/2$ and an upper level with $J' = 3/2$ that moves through a standing wave formed by two counter-propagating laser beams with orthogonal linear polarizations (e.g. along $\hat{\mathbf{e}}_x$ and $\hat{\mathbf{e}}_y$). The resultant polarization depends on the relative phase of the two laser beams and varies with position, as shown in Fig. 9.18(b). This polarization gradient causes the periodic modulation of the light shift of the states in the lower level. The strength of the interaction with the light depends on M_J and $M_{J'}$ in the lower and upper levels, respectively. To understand this in detail, consider

Fig. 9.17 The laser cooling mechanism in a standing wave with a spatially-varying polarization. The energy levels of the atom are perturbed by the light in a periodic way, so that the atoms travel up and down hills and valleys (maxima and minima) in the potential energy. Kinetic energy is lost when the atom absorbs laser light at the top of a hill and emits a spontaneous photon of higher frequency, so that it ends up in a valley. This has been called the Sisyphus effect and can be made more probable than the reverse process, so that there is strong laser cooling. Thus atoms in a standing wave are cooled below the Doppler cooling limit (the lowest temperature achievable with scattering force alone).



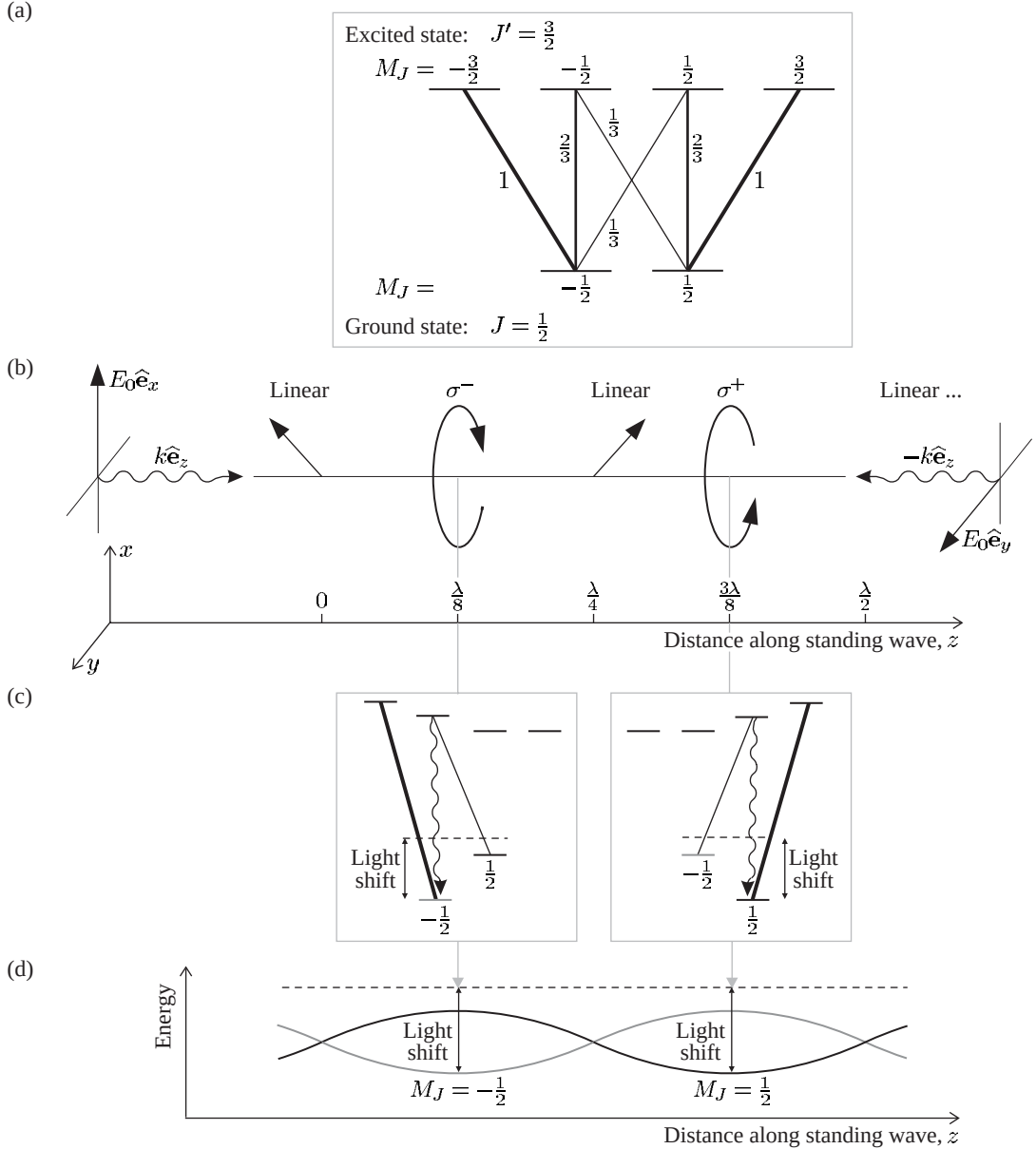
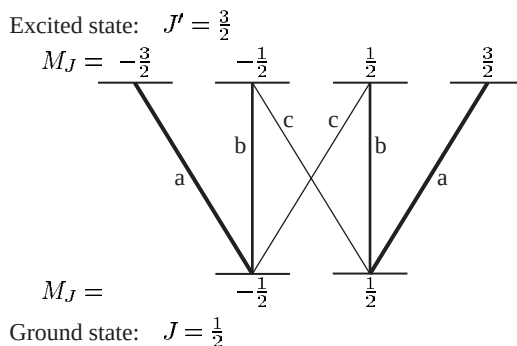


Fig. 9.18 Details of the Sisyphus cooling mechanism. (a) The electric dipole transitions between two levels with angular momenta $J = 1/2$ and $J' = 3/2$. The relative strength of each transition is indicated—this gives the relative intensity when the states in the upper level are equally populated (each state has the same radiative lifetime). (b) The polarization in a standing wave formed by two laser beams that propagate along $\hat{e}_z$ and $-\hat{e}_z$, and have orthogonal linear polarizations along $\hat{e}_x$ and $\hat{e}_y$, respectively. The resultant electric field is circularly polarized $(\hat{e}_x \pm i\hat{e}_y)/\sqrt{2}$ at positions where the two counter-propagating beams have a phase difference of $\pm\pi/2$. The polarization changes from σ^+ to σ^- over a distance of $\Delta z = \lambda/4$, and between these positions the light has elliptical or linear polarization. (c) The energies of the states at positions of σ^- and σ^+ polarization (the unperturbed energy of the lower level is shown as a dotted line). Absorption of the circularly-polarized light followed by spontaneous emission transfers the population into the state with lowest energy (largest light shift). (d) The light shift varies with position and the optical pumping process, outlined in (c), transfers atoms from the top of a hill to the bottom of a valley (as shown in Fig. 9.17); or at least this process in which atoms lose energy happens more often than the other way around.

Fig. 9.19 The intensities of the components of the $J = 1/2$ to $J = 3/2$ transition are represented by a , b and c (as in Example 7.3) and their relative values can be determined from sum rules. The sum of the intensities from each of the upper states is the same: $a = b + c$; because normally the lifetime of an atom does not depend on its orientation. (A similar rule applies to the states in the lower level but this does not yield any further information in this case.) When the states of the upper level are equally populated the atom emits unpolarized radiation; hence $a + c = 2b$. We now have two simultaneous equations whose solution is $b = \frac{2}{3}a$ and $c = \frac{1}{3}a$, so the relative intensities are $a : b : c = 3 : 2 : 1$.



⁵⁵This is the same convention for describing polarization that we used for the magneto-optical trap; σ^+ and σ^- refer to transitions that the radiation excites in the atom, $\Delta M_J = \pm 1$, respectively. In laser cooling we are mainly interested in determining what transitions occur, and this depends on the sense of rotation of the electric field around the quantization axis of the atom (as described in Section 2.2). As stated previously, the electric field of the radiation drives the bound atomic electron(s) around in the same sense as the electric field; the circularly-polarized radiation travelling parallel to the quantization axis that is labelled σ^+ imparts positive angular momentum to the atom. The handedness of the polarization can be deduced from this statement for a given direction of propagation, if necessary.

a position where the light has σ^+ polarization;⁵⁵ here the interaction for the $M_J = 1/2$ to $M_{J'} = 3/2$ transition is stronger than that for the $M_J = -1/2$ to $M_{J'} = 1/2$. (The squares of the Clebsch–Gordan coefficients are $2/3$ and $1/3$, respectively, for these two transitions, as determined from the sum rules as shown in Fig. 9.19). For light with a frequency detuning to the red ($\delta < 0$), both of the M_J states in the lower level ($J = 1/2$) are shifted downwards; the $M_J = +1/2$ state is shifted to a lower energy than the $M_J = -1/2$ state. Conversely, at a position of σ^- polarization the $M_J = -1/2$ state is lower than the $M_J = 1/2$ state. The polarization changes from σ^- to σ^+ over a distance of $\Delta z = \lambda/4$, so that the light shift varies along the standing wave, as shown in Fig. 9.18(d). An atom moving over these hills and valleys in the potential energy speeds up and slows down as kinetic and potential energy interchange, but its total energy does not change if it stays in the same state.

To cool the atom there must be a mechanism for dissipating energy and this occurs through absorption and spontaneous emission—the process in which an atom absorbs light at the top of a hill and then decays spontaneously back down to the bottom of a valley has a higher probability than the reverse process. Thus the kinetic energy that the atom converts into potential energy in climbing the hill is lost (taken away by the spontaneously-emitted photon); the atom ends up moving more slowly, at the bottom of a valley. This was dubbed the ‘Sisyphus’ effect after a character in Greek mythology who was condemned by the gods

to repeatedly roll a stone to the top of a hill.⁵⁶

To explain the transfer between the M_J states of the lower level let us again consider the details of what happens at a particular position where the light has σ^+ polarization (see Fig. 9.18(d)). Absorption of σ^+ light excites an atom from the state $M_J = -1/2$ to $M_{J'} = 1/2$. An atom in this excited state may decay to either of the lower M_J states; if it returns to $M_J = -1/2$ then the process restarts, but it may go into the $M_J = +1/2$ state from which it cannot return (because σ^+ light excites the transition from $M_J = 1/2$ to $M_{J'} = 3/2$ and the excited state of this transition only decays to $M_J = +1/2$). Thus the σ^+ light results in a one-way transfer $M_J = -1/2$ to $M_J = +1/2$ (via an excited state). This process in which absorption of light transfers population into a given state is known as **optical pumping**.⁵⁷ In Sisyphus cooling the optical pumping at a position of σ^+ polarization takes an atom from the top of a hill, in the potential energy for the $M_J = -1/2$ state, and transfers it down into a valley of the potential energy for the $M_J = 1/2$ state. The atom continues its journey in the $M_J = 1/2$ state until it gets optically pumped back to $M_J = -1/2$ at a position of σ^- polarization.⁵⁸ In each transfer the atom loses an energy U_0 approximately equal to the height of the hills (relative to the bottom of the valleys). This energy is roughly equal to the light shift in eqn 9.46.⁵⁹

This physical picture can be used to estimate the rates of cooling and heating in the Sisyphus mechanism—the balance between these gives the equilibrium temperature (cf. the treatment of Doppler cooling in Section 9.3).⁶⁰ Such a treatment shows that atoms in a standing wave have a typical kinetic energy $\sim U_0$. This suggests that the Sisyphus mechanism works until atoms can no longer climb the hills and remain stuck in a valley (cf. an optical lattice). This simple picture predicts that the temperature is related to the intensity and frequency detuning by

$$k_B T \simeq U_0 \propto \frac{I}{|\delta|}, \quad (9.53)$$

which is borne out by more detailed calculation.

9.7.3 Limit of the Sisyphus cooling mechanism

In a typical optical molasses experiment there are the following two stages. Initially, the laser beams have a frequency several line widths below the atomic resonance ($\delta \sim -\Gamma$) and intensities $\sim I_{\text{sat}}$ to give a strong radiation force. Then the laser frequency is changed to be further from resonance (and the intensity may be reduced as well) to cool the atoms to lower temperatures below the Doppler limit. The initial stage of Doppler cooling, as described in Section 9.3, is essential because the sub-Doppler cooling mechanisms only operate over a very narrow range

⁵⁶In addition to Sisyphus cooling, Dalibard and Cohen-Tannoudji found another sub-Doppler cooling mechanism called **motion-induced orientation**. This mechanism leads to an imbalance in scattering from counter-propagating beams that is much more sensitive to velocity, and hence produces a stronger damping than the imbalance caused by the Doppler effect in the ‘ordinary’ optical molasses technique. In a standing wave made from beams of opposite circular polarization (σ^+ to σ^-), a stationary atom has the population distributed over the magnetic sub-levels of the ground state in a symmetrical way, so that there is equal scattering from each beam and no net force. An atom moving through a gradient of polarization, however, sees a changing direction of the electric field and this causes a change in the distribution over the sub-levels (orientation by optical pumping) leading to a difference in the probability of absorption from each beam. In real optical molasses experiments, the three mutually-orthogonal pairs of laser beams create a complex three-dimensional pattern of polarization and a combination of sub-Doppler cooling mechanisms takes place.

⁵⁷Optical pumping in atomic vapours at room temperature was used for very precise radio-frequency spectroscopy even before the laser was invented, e.g. to measure the splitting between the Zeeman sub-levels as described in Thorne (1999) and Corney (2000).

⁵⁸The atom may travel over many hills and valleys between excitations, and the absorption and emission does not always remove energy, but averaged over many events this stochastic process dissipates energy.

⁵⁹Actually, it is about two-thirds of the light shift for the case shown in Fig. 9.17.

⁶⁰The heating arises from fluctuations in the dipole force—the direction of this force changes as an atom jumps from one M_J state to another. See Metcalf and van der Straten (1999) for a quantitative treatment.

⁶¹Broadly speaking, in Sisyphus cooling the force averages to zero for atoms that travel over many hills and valleys in an optical-pumping time. Thus this mechanism works for velocities v such that $v\tau_{\text{pump}} \lesssim \lambda/2$. This velocity range is less than the capture velocity for Doppler cooling by the ratio τ/τ_{pump} .

⁶²This assumes that each degree of freedom has energy

$$\frac{1}{2}k_{\text{B}}T_{\text{r}} = E_{\text{r}}. \quad (9.54)$$

⁶³Heavy alkalis such as Cs and Rb have a very low recoil limit and these elements can be laser cooled to a few μK . Such temperatures can only be achieved in practice when stray magnetic fields that would perturb the M_F states are carefully controlled—a Zeeman shift $\mu_{\text{B}}B$ comparable with the light shift U_0 will affect the Sisyphus cooling mechanism, i.e. if $\mu_{\text{B}}B \sim U_0$. For $U_0/k_{\text{B}} = 3\mu\text{K}$ this criterion implies that $B < 5 \times 10^{-5}\text{ T}$, which is an order of magnitude less than the Earth's magnetic field ($5 \times 10^{-4}\text{ T}$).

⁶⁴The assumption that the distribution has a Gaussian shape becomes worse at the lowest velocities of only a few times the recoil velocity—the smallest amount by which the velocity can change. Commonly, the distribution develops a sharp peak around $v = 0$ with wide wings. In such cases the full distribution needs to be specified, rather than a single parameter such as the root-mean-square velocity, and the notion of a ‘temperature’ may be misleading. This remark is even more relevant for cooling below the recoil limit, as described in the following section.

⁶⁵In a high-intensity standing wave, a combination of the dipole force and spontaneous scattering dissipates the energy of a two-level atom, as shown by Dalibard and Cohen-Tannoudji (1985). This high-intensity Sisyphus mechanism damps the atomic motion for a frequency detuning to the blue (and the opposite for the low-intensity effect), and the hills and valleys in the potential energy arise directly from the variation in intensity, as in an optical lattice, rather than a gradient of polarization.

of velocities.⁶¹

The equilibrium temperature in sub-Doppler cooling does not decrease indefinitely in proportion to $I/|\delta|$. Sisyphus cooling stops working when the loss in energy in going from the top of a hill (in the potential energy) to the valley bottom is balanced by the recoil energy acquired in spontaneous emission, $U_0 \simeq E_{\text{r}}$. For this condition there is no net loss of energy in optical pumping between M_J states. Thus the lowest temperatures achieved are equivalent to a few times the recoil energy, $T \simeq E_{\text{r}}/k_{\text{B}}$. At this recoil limit⁶² the temperature is given by

$$k_{\text{B}}T_{\text{r}} = \frac{\hbar^2 k^2}{M} \equiv \frac{\hbar^2}{M\lambda^2}. \quad (9.55)$$

For sodium the temperature at the recoil limit is only $2.4\mu\text{K}$ (cf. $T_{\text{D}} = 240\mu\text{K}$). Typically, the optical molasses technique can reach temperatures that are an order of magnitude above the recoil limit, but still well below the Doppler cooling limit.⁶³

The meaning of temperature must be considered carefully for dilute gas clouds. In a normal gas at room temperature and pressure the interatomic collisions establish thermal equilibrium and give a Maxwell–Boltzmann distribution of velocities. A similar Gaussian distribution is often obtained in laser cooling, where each atom interacts with the radiation field independently (for moderate densities, as in the optical molasses technique) and an equivalent temperature can be assigned that characterises the width of this distribution (see eqn 8.3).⁶⁴ From the quantum point of view, the de Broglie wavelength of the atom is more significant than the temperature. At the recoil limit the de Broglie wavelength roughly equals the wavelength of the cooling radiation, $\lambda_{\text{dB}} \sim \lambda_{\text{light}}$, because the atomic momentum equals that of the photons (and for both, $\lambda = h/p$ is the relationship between wavelength and momentum p).

This section has described the Sisyphus cooling that arises through a combination of the spatially-varying dipole potential, produced by the polarization gradients, and optical pumping. It is a subtle mechanism and the beautifully-detailed physical explanation was developed in response to experimental observations. It was surprising that the small light shift in a low-intensity standing wave has any influence on the atoms.⁶⁵ The recoil limit is an important landmark in laser cooling and the next section describes a method that has been invented to cool atoms below this limit.

9.8 Raman transitions

9.8.1 Velocity selection by Raman transitions

Raman transitions involve the simultaneous absorption and stimulated emission by an atom. This process has many similarities with the two-photon transition described in Section 8.4 (see Appendix E). A coherent Raman transition between two levels with an energy difference of $\hbar\omega_{12}$

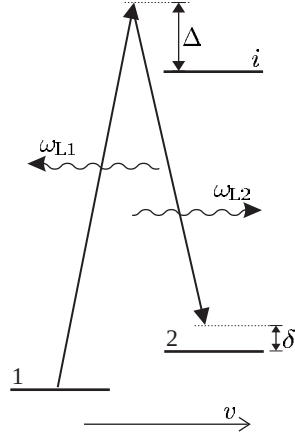


Fig. 9.20 A Raman transition between levels 1 and 2 driven by two laser beams of (angular) frequencies ω_{L1} and ω_{L2} . For a resonant Raman process the frequency detuning $\delta \simeq 0$, and the detuning Δ from the intermediate state remains large, so that excitation by single-photon absorption is negligible in comparison to the coherent transfer from $|1\rangle$ to $|2\rangle$. In this example the atom has velocity v along the direction of the laser beam with frequency ω_{L2} , and the laser beam with frequency ω_{L1} propagates in the opposite direction.

is illustrated in Fig. 9.20. For two beams of frequencies ω_{L1} and ω_{L2} the condition for resonant excitation is

$$\omega_{L1} + k_1 v - (\omega_{L2} - k_2 v) = \omega_{L1} - \omega_{L2} + \frac{v}{c} (\omega_{L1} + \omega_{L2}) = \omega_{12}. \quad (9.56)$$

For counter-propagating beams the Doppler shifts add to make the Raman transition sensitive to the velocity—about twice as sensitive as a single-photon transition.⁶⁶ Direct excitation of the transition by radio-frequency radiation, or microwaves, at angular frequency ω_{12} is insensitive to the motion. The great advantage of the Raman technique for velocity selection (and cooling) arises from its extremely narrow line width (comparable with that of radio-frequency methods) of Raman transitions between levels that have long lifetimes, e.g. hyperfine levels in the ground configuration of atoms for which spontaneous decay is negligible. To fully exploit the advantage of this narrow line width, the difference in frequency between the two laser beams $\Delta\omega = \omega_{L1} - \omega_{L2}$ must be controlled very precisely. This can be achieved by taking two independent lasers and implementing sophisticated electronic servo-control of the frequency difference between them, but it is technically easier to pass a single laser beam through a phase modulator—the resultant frequency spectrum contains ‘sidebands’ whose difference from the original laser frequency equals the applied modulation frequency from a microwave source.⁶⁷ The selected velocity v is determined by

$$2kv = \omega_{12} - (\omega_{L1} - \omega_{L2}), \quad (9.57)$$

where $k = (\omega_{L1} + \omega_{L2})/c$ is the mean wavevector.

Raman transitions between levels with negligible broadening from spontaneous decay or collisions have a line width determined by the interaction time: for a pulse of duration τ_{pulse} the Fourier transform limit gives⁶⁸

$$\frac{\Delta\nu}{\lambda} \simeq \frac{1}{\tau_{\text{pulse}}}. \quad (9.58)$$

For a visible transition with a wavelength of 600 nm ⁶⁹ a pulse of duration

⁶⁶In contrast, two counter-propagating laser beams of the same frequency give Doppler-free two-photon spectra:

$$\omega_L + kv + (\omega_L - kv) = 2\omega_L,$$

as in eqn 8.20. If a two-photon transition is excited by two laser beams with different frequencies then the Doppler shifts do not cancel exactly.

⁶⁷For a laser beam with (angular) frequency ω , phase modulation at frequency Ω leads to a spectrum containing the frequencies $\omega \pm n\Omega$, with n integer. This can be used to carry out Raman excitation, e.g. with $\omega_{L1} = \omega$ and $\omega_{L2} = \omega - \Omega$.

⁶⁸Similar to that for the single-photon transition in Section 7.1.2.

⁶⁹As in the case of sodium that was used in the first Raman experiments with cold atoms. In sodium, levels 1 and 2 are the hyperfine levels with $F = 1$ and 2 of the $3s$ configuration, and the intermediate level i is $3p \ ^2P_{3/2}$.

⁷⁰A useful comparison can be made with the method of reducing the Doppler broadening shown in Fig. 8.2, in which a narrow slit is used to collimate an atomic beam and so reduce the spread of transverse velocities.

$\tau_{\text{pulse}} = 600 \mu\text{s}$ selectively transfers atoms in a range of width $\Delta v \simeq 1 \text{ mm s}^{-1}$. This is about thirty times less than v_r and equivalent to a ‘temperature’ of $T_r/900$ for the motion along the axis of the laser beams. This velocity selection does *not* produce any more cold atoms than at the start—it just separates the cold atoms from the others—so it has a different nature to the laser cooling processes described in the previous sections.⁷⁰

9.8.2 Raman cooling

The previous section showed that Raman transitions give the precision of radio-frequency spectroscopy combined with a sensitivity to the Doppler shift twice that of single-photon (optical) transitions. Raman cooling exploits the extremely high velocity resolution of coherent Raman transitions to cool atoms below the recoil limit. The complete sequence of operations in Raman cooling is too lengthy to describe here, but the important principle can be understood by considering how atoms with a velocity distribution that is already below the recoil limit are cooled further. Figure 9.21 shows such an initial distribution in level 1 (the lower hyperfine level in the ground configuration of the atom; level 2 is the upper hyperfine level). Raman cooling uses the following steps.

- (a) Velocity selection by a Raman pulse that transfers atoms with velocities in the range from $v - \Delta v/2$ to $v + \Delta v/2$ up to level 2, where they have velocities centred about $v - 2v_r$. (The process of absorption and stimulated emission in the opposite direction changes the atom’s velocity by $2v_r$.⁷¹)
- (b) Atoms in level 2 are excited to level i by another laser beam and can decay back to level 1 with velocities centred around $v - v_r$ (including the change in velocity produced by absorption). Spontaneous emission goes in all directions so that the atoms return to level 1 with velocities anywhere in the range v to $v - 2v_r$.⁷²

⁷¹Equivalently, the atom’s momentum changes by $\hbar\mathbf{k}_1 - \hbar\mathbf{k}_2 \simeq 2\hbar\mathbf{k}$. (Here $\omega_{12} \ll \omega_1 \simeq \omega_2$.)

⁷²Some atoms fall back into level 2 and are excited again until eventually they end up in level 1. Atoms that undergo more than one excitation receive additional impulses from the absorbed and emitted photons, which reduces the cooling efficiency but does not affect the principle.

⁷³For velocity selection of atoms with $v < 0$ the direction of the beams is reversed so that these atoms are distributed into the range v to $v+2v_r$ (that includes $v = 0$).

It might appear that this cycle of a velocity-selective Raman pulse followed by repumping has made things worse since the final spread of velocities is comparable to, if not greater than the initial spread. Crucially, however, some atoms fall back into level 1 with velocities very close to zero so the number of very slow atoms has increased, and increases further for each repetition of the cycle with different initial velocity.⁷³ Precise control of the Raman pulses ensures that atoms with velocities in the narrow range $-\delta v < v < \delta v$ are never excited, so that after many cycles a significant fraction of the population accumulates in this narrow velocity class with $\delta v \ll v_r$. In this Raman cooling process the atomic velocity undergoes a random walk until either it falls into the desired velocity class around $v = 0$, and remains there, or diffuses away to higher velocities. The recoil limit is circumvented because atoms with $v \simeq 0$ do not interact with the light and this sub-recoil cooling mechanism does not involve a radiation force (in contrast to the Doppler and sub-Doppler cooling mechanisms described in previous sections).

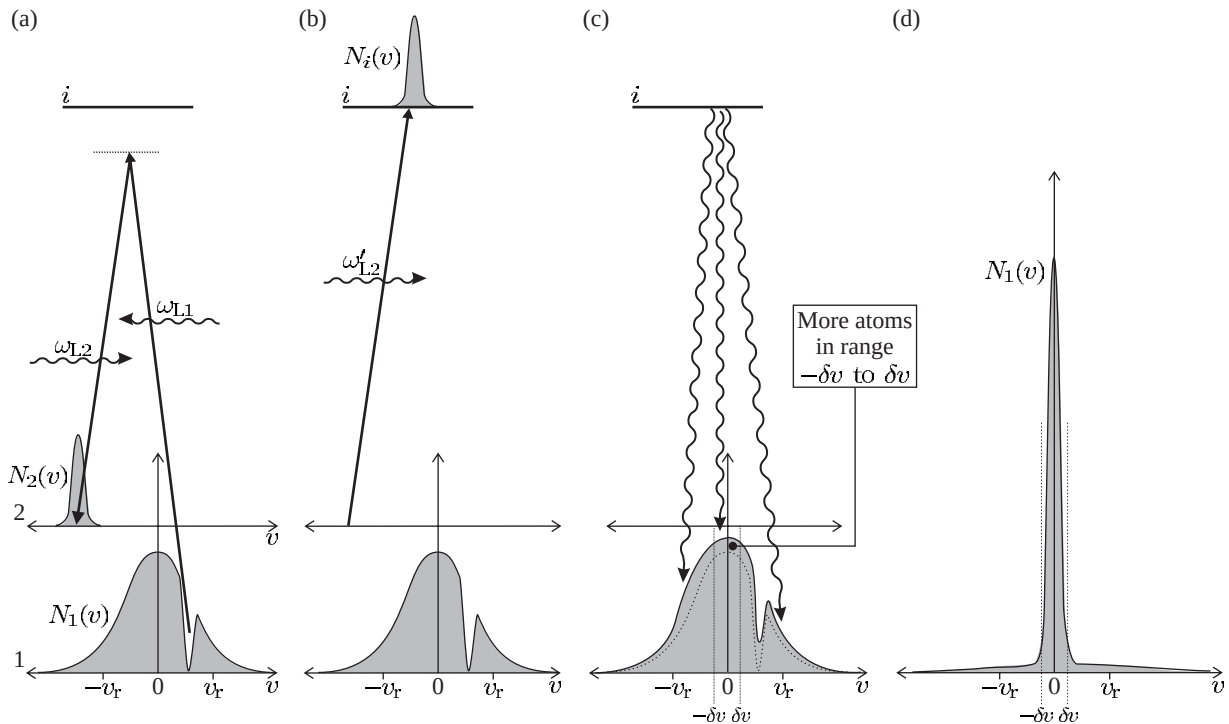


Fig. 9.21 One step in the sequence of operations in Raman cooling. (a) Velocity selection by a Raman pulse that transfers atoms that have velocities within a certain narrow range from $|1\rangle$ to $|2\rangle$ —the process of absorption and stimulated emission changes the atomic velocity by $-2v_r$. (b) Atoms are excited from level 2 to level i by another laser beam—in this process the atomic velocity changes by v_r . (c) Atoms decay to level 1 by spontaneous emission—the recoil in a random direction means that the atoms return to level 1 with a component of velocity anywhere in the range v to $v - 2v_r$, where v is the initial velocity. There are more atoms in the narrow velocity class around $v = 0$ than at the start of the sequence. (d) Repetition of the sequence with different initial velocities increases the number of atoms with $v \simeq 0$ until they are ‘piled’ up in a distribution whose width is much less than the recoil velocity v_r .

The time taken for atoms to fall (randomly) into a velocity class of width $2\delta v$ increases as δv decreases and this determines the final velocity spread achievable by Raman cooling in practice.

Raman cooling works well in one dimension, but it is much less efficient in three dimensions where the target is to have all three components v_x , v_y and v_z between $\pm\delta v$. Another method of sub-recoil cooling called velocity-selective coherent population trapping is also a stochastic process, see Metcalf and van der Straten (1999) and Bardou *et al.* (1991). Raman transitions are also used for matter-wave interferometry based on ultra-cold atoms (Chapter 10).

9.9 An atomic fountain

The slow atoms produced by laser cooling have led to a dramatic improvement in measurements whose resolution is limited by the interaction time. Cold atoms can be confined in dipole-force traps⁷⁴ for long

⁷⁴Or magnetic traps as described in the next chapter.

⁷⁵In principle, the perturbation can be calculated and corrected for, but without perfect knowledge of the trapping potential this leaves a large uncertainty. There are currently proposals for frequency standards based on transitions in optically-trapped atoms for which the light shift cancels out, i.e. the lower and upper levels of the narrow transition have very similar light shifts.

⁷⁶In such an atomic fountain the atoms have an initial velocity of $v_z = (2gh)^{1/2} = 4 \text{ m s}^{-1}$.

⁷⁷Caesium has a resonance wavelength of 852 nm and a relative atomic mass of 133.

⁷⁸Only the spread in the radial direction leads to a loss of atoms, so velocity selection in two dimensions by Raman transitions, or otherwise, is useful.

⁷⁹At the extraordinary precision of these experiments, collisions between ultra-cold caesium atoms cause an observable frequency shift of the hyperfine transition (proportional to the density of the atoms). Therefore it is undesirable for this density to change during the measurement.

periods of time; however, the trapping potential strongly perturbs the atomic energy levels and hinders accurate measurements of the transition frequencies.⁷⁵ The highest-accuracy measurements use atoms in free fall, as shown in Fig. 9.16. This apparatus launches cold atoms upwards with velocities of a few m s^{-1} , so that they travel upwards for a short distance before turning around and falling back down under gravity—this forms an atomic fountain.

A particularly important use of atomic fountains is to determine the frequency of the hyperfine-structure splitting in the ground configuration of caesium since this is used as the primary standard of time. Each atom passes through a microwave cavity on the way up and again on its way down, and these two interactions separated in time by T lead to Ramsey fringes (Fig. 7.3) with frequency width $\Delta f = 1/(2T)$, as described in Section 7.4. Simple Newtonian mechanics shows that a fountain of height $h = 1 \text{ m}$ gives $T = 2(2h/g)^{1/2} \simeq 1 \text{ s}$, where g is the gravitational acceleration.⁷⁶ This is several orders of magnitude longer than the interaction time for a thermal atomic beam of caesium atoms (Section 6.4.2). This is because the measurement time on Earth is limited by gravity, and an obvious, but not simple, way to obtain further improvement is to put an apparatus into space, e.g. aboard a satellite or space station in orbit. Such an apparatus has the same components as an atomic fountain, but the atoms only pass once through the microwave interaction region and are detected on the other side—pushing the atoms gently so that they move very slowly through the microwave cavity gives measurement times exceeding 10 s.

Cold atoms obtained by laser cooling are essential for both the atomic fountain and atomic clocks in space, as shown by the following estimate for the case of a fountain. The entrance and exit holes of the microwave cavity have a diameter of about 1 cm. If the atoms that pass through the cavity on the way up have a velocity spread about equal to the recoil velocity $v_r = 3.5 \text{ mm s}^{-1}$ for caesium,⁷⁷ then the cloud will have expanded by $\sim 4 \text{ mm}$ by the time it falls back through the cavity.⁷⁸ Thus a reasonable fraction of these atoms, that have a temperature close to the recoil limit, pass back through the cavity and continue down to the detection region. Clearly, for a 10 s measurement time the effective temperature of the cloud needs to be well below the recoil limit.⁷⁹ These general considerations show the importance of laser cooling for the operation of an atomic fountain. Some further technical details are given below.

The atoms are launched upwards by the so-called ‘moving molasses’ technique, in which the horizontal beams in the six-beam configuration shown in Fig. 9.5 have angular frequency ω , and the upward and downward beams have frequencies $\omega + \Delta\omega$ and $\omega - \Delta\omega$, respectively. In a reference frame moving upwards with velocity $\mathbf{v} = (\Delta\omega/k)\hat{\mathbf{e}}_z$ the Doppler shift is $\Delta\omega$, so that all the beams appear to have the same frequency. Therefore the optical molasses mechanisms damp the atomic velocity to zero with respect to this moving frame. These atoms have the same velocity spread about their mean velocity as atoms in the optical molasses technique with a stationary light field ($\Delta\omega = 0$), so the

temperature is the same in both cases.⁸⁰

In an atomic fountain the scheme for detecting that a microwave transition has occurred is very different to that in an atomic beam (Section 6.4). The ground configuration of caesium has $J = 1/2$ (like all alkalis) and the two hyperfine levels are $F = 3$ and 4 (for the only stable isotope that has nuclear spin $I = 7/2$). If the atoms start in the lower level $F = 3$ then the microwave radiation transfers a fraction of the atoms to the $F = 4$ level. This fraction is determined when the atoms fall through a laser beam that detects atoms in the $F = 4$ level, by exciting a transition from this level and monitoring the fluorescence, see Fig. 9.16. (Atoms in the $F = 3$ level pass through undetected.⁸¹) Figure 7.3 shows a plot of the transition probability between the hyperfine levels as a function of the microwave frequency—so-called Ramsey fringes. The narrow line width means that the frequency of the microwave source used to drive the transition can be set very precisely to the caesium hyperfine frequency. Such an apparatus maintains the frequency of the microwave source stable to better than 1 part in 10^{15} , or 32 ns per year. Many causes of perturbations that might give frequency shifts are small because of the atoms' low atomic velocity, but the Zeeman effect of magnetic fields remains a limitation. Experiments use the $F = 3$, $M_F = 0$ to $F = 4$, $M_F = 0$ transition because states with $M_F = 0$ have no first-order Zeeman shift. Nowadays, such caesium fountain frequency standards play an important role in guiding the ensemble of clocks in national standards laboratories around the world that give agreed Universal Time.⁸²

⁸⁰These caesium atoms have a velocity spread of about $3v_T \simeq 10 \text{ mm s}^{-1}$. These atoms could be used directly if the measurement time is 0.3 s, but there would be a large loss of atoms in a higher fountain with $T = 1$ s.

⁸¹To normalise the signal the atoms in the $F = 3$ level are detected with a second probe laser beam (not shown in the figure).

⁸²The important uses of such clocks were given in Section 6.4.2 and up-to-date information can be found on the web sites of national standards laboratories.

9.10 Conclusions

The techniques that have been developed to reduce the temperature of atoms from 1000 K to well below $1 \mu\text{K}$ have had an enormous impact on atomic physics. Laser cooling has made it possible to manipulate neutral atoms in completely new ways and to trap them by magnetic and dipole forces. Some important applications of atom trapping have been mentioned, such as the great improvement in precision measurements, and others are given in later chapters, e.g. Bose–Einstein condensation and the laser cooling of trapped ions.

The important principles of radiation forces have been discussed, namely: the way in which the scattering force dissipates the energy of atoms and cools them to the Doppler cooling limit; the trapping of atoms by the dipole force in various configurations including optical lattices; and sub-Doppler cooling by the Sisyphus mechanism and sub-recoil cooling. This chapter greatly simplifies the real story of laser cooling for the sake of a clear presentation; the book *Laser cooling and trapping of atoms* by Metcalf and van der Straten (1999) gives a more comprehensive description of the important contributions made by many people and many references to other material—see also the review by Wieman

et al. (1999). Various internet resources and popular descriptions can be found on the web site of the Nobel foundation.

Exercises

More advanced problems are indicated by a *.

(9.1) *Radiation pressure*

What force does radiation exert on the head of a person wearing a black hat of radius 15 cm when the sun is directly overhead. Estimate the ratio of this radiation force to the weight of the hat.

(9.2) *An argument for photon momentum (due to Enrico Fermi)*

An atom moving at velocity v absorbs a photon propagating in the opposite direction (as in Fig. 9.1). In the laboratory frame of reference the photon has (angular) frequency ω and momentum q_{ph} . In the rest frame of the atom the photon has (angular) frequency ω_0 , where $\hbar\omega_0 = E_2 - E_1$ is the energy of the (narrow line width) transition between levels 1 and 2. After the absorption the system has a total energy of $\frac{1}{2}M(v - \Delta v)^2 + \hbar\omega_0$.

- Write down the equations for conservation of energy and momentum.
- Expand the equation for conservation of energy, neglecting the term of order $(\Delta v)^2$. (The change in velocity Δv is small compared to v .)
- Use the usual expression for the fractional Doppler shift $(\omega - \omega_0)/\omega = v/c$ to find an expression for the photon momentum q_{ph} .

(9.3) *Heating from photon recoil*

This exercise is based on a treatment of laser cooling by Wineland and Itano (1979). The angular frequencies of radiation absorbed and emitted by an atom are given by

$$\omega_{\text{abs}} = \omega_0 + \mathbf{k}_{\text{abs}} \cdot \mathbf{v} - \frac{1}{2}\omega_0 \left(\frac{v}{c}\right)^2 + \frac{E_r}{\hbar},$$

$$\omega_{\text{em}} = \omega_0 + \mathbf{k}_{\text{em}} \cdot \mathbf{v}' - \frac{1}{2}\omega_0 \left(\frac{v'}{c}\right)^2 - \frac{E_r}{\hbar},$$

where $|\mathbf{k}_{\text{abs}}| = \omega_{\text{abs}}/c$ and $|\mathbf{k}_{\text{em}}| = \omega_{\text{em}}/c$ are the wavevectors of the absorbed and emitted photons, respectively, $\mathbf{v}$ is the velocity of the atom before the photon is absorbed, and similarly $\mathbf{v}'$ is the velocity of the atom before emission. Prove these results from conservation of (relativistic) energy and

momentum (keeping terms of order $(v/c)^2$ in the atomic velocity and $E_r/\hbar\omega_0$ in the recoil energy). Averaged over many cycles of absorption and emission, the kinetic energy of the atom changes by

$$\Delta E_{\text{ke}} = \hbar(\omega_{\text{abs}} - \omega_{\text{em}}) = \hbar\mathbf{k}_{\text{abs}} \cdot \mathbf{v} + 2E_r$$

for each scattering event. Show that this result follows from the above equations with certain assumptions, that should be stated. Show that, when multiplied by the scattering rate R_{scatt} , the terms $\hbar\mathbf{k}_{\text{abs}} \cdot \mathbf{v}$ and $2E_r$ give cooling and heating at comparable rates to those derived in the text for the optical molasses technique.

(9.4) *The angular momentum of light*

An atom in a 1S_0 level is excited to a state with $L = 1$, $M_L = 1$ by the absorption of a photon (a σ^+ transition). What is the change in the atomic angular momentum?

A laser beam with a power of 1 W and a wavelength of 600 nm passes through a waveplate that changes the polarization of the light from linear to circular. What torque does the radiation exert on the waveplate?

(9.5) *Slowing H and Cs with radiation*

Atomic beams of hydrogen and caesium are produced by sources at 300 K and slowed by counter-propagating laser radiation. In both cases calculate (a) the stopping distance at half of the maximum deceleration, and (b) compare the Doppler shift at the initial velocity with the natural width of the transition. (Data are given in Table 9.1.)

(9.6) *The Doppler cooling and recoil limits*

Calculate the ratio T_D/T_r for rubidium (from eqns 9.28, 9.55 and the data in Table 9.1).

(9.7) *Damping in the optical molasses technique*

- For the particular case of a frequency detuning of $\delta = -\Gamma/2$ the slope of the force versus velocity curve, shown in Fig. 9.6, at $v = 0$ equals the peak force divided by $\Gamma/(2k)$. Use this to

estimate $\partial F/\partial v$ and hence to determine the damping coefficient α for an atom in a pair of counter-propagating laser beams, under these conditions.

- (b) Estimate the damping time for a sodium atom in the optical molasses technique when each laser beam has intensity I_{sat} and $\delta = -\Gamma/2$.

(9.8) *Laser cooling of a trapped ion*

A trapped Ca^+ ion undergoes simple harmonic motion with an oscillation frequency of $\Omega = 2\pi \times 100 \text{ kHz}$. The ion experiences a radiation force from laser light of wavelength 393 nm and intensity I that excites a transition with $\Gamma = 2\pi \times 23 \times 10^6 \text{ s}^{-1}$. The frequency detuning δ does not depend on the ion's position within the trap.

- (a) Show that the force on the ion has the form $F = -\kappa(z - z_0) - \alpha v$. Describe the ion's motion.
- (b) Find the static displacement z_0 of the ion from the centre of the harmonic potential, along the direction of the laser beam, for $\delta = -\Gamma/2$ and $I = 2I_{\text{sat}}$.
- (c) Show that, to a good approximation, the damping coefficient can be written in the form

$$\alpha \propto \frac{xy}{(1 + y + x^2)^2}, \quad (9.59)$$

where the variables x and y are proportional to δ and I , respectively. Maximise this function of two variables and hence determine the intensity and frequency detuning that give the maximum value of α .

- (d) The kinetic energy of small oscillations about z_0 decays with a damping time of $\tau_{\text{damp}} = M/\alpha$. Show that this damping time is inversely proportional to the recoil energy.⁸³ Evaluate this minimum value of τ_{damp} for a calcium ion of mass $M \simeq 40 \text{ a.m.u.}$

Comment. This treatment of Doppler cooling for a single laser beam is accurate for any intensity (even above I_{sat}), whereas the approximation that two laser beams (as in the optical molasses technique) give twice as much damping as a single beam is not accurate at high intensities.

(9.9) *The properties of a magneto-optical trap*

- (a) Obtain an expression for the damping coefficient α for an atom in two counter-propagating laser beams (each of intensity I),

taking into account saturation. (Use the results of the previous exercise with the modification $I \rightarrow 2I$ in both the numerator and denominator, or otherwise.) Determine the minimum damping time (defined in eqn 9.19) of a rubidium atom in the optical molasses technique (with two laser beams).

- (b) The force on an atom in an MOT is given by eqn 9.30. Assume the worst-case scenario in the calculation of the damping and the restoring force, along a particular direction, i.e. that the radiation force arises from two counter-propagating laser beams (each of intensity I) but the saturation of the scattering rate depends on the total intensity $6I$ of all six laser beams. Show that the damping coefficient can be written in the form

$$\alpha \propto \frac{xy}{(1 + y + x^2)^2},$$

where $x = 2\delta/\Gamma$ and $y = 6I/I_{\text{sat}}$. Using the results of the previous exercise, or otherwise, determine the nature of the motion for a rubidium atom in an MOT with the values of I and δ that give maximum damping, and a field gradient 0.1 T m^{-1} (in the direction considered).

(9.10) *Zeeman slowing in a magneto-optical trap*

- (a) Instead of the optimum magnetic field profile given in eqn 9.11, a particular apparatus to slow sodium atoms uses a linear ramp

$$B(z) = B_0 \left(1 - \frac{z}{L}\right)$$

for $0 \leq z \leq L$, and $B(z) = 0$ outside this range. Explain why a suitable value for B_0 is the same as in eqn 9.12. Show that the minimum value of L is $2L_0$, where L_0 is the stopping distance for the optimum profile.

- (b) The capture of atoms by a magneto-optical trap can be considered as Zeeman slowing in a uniform magnetic field gradient, as in part (a). In this situation the maximum velocity captured by an MOT with laser beams of radius 0.5 cm is equivalent to the velocity of atoms that come to rest in a distance of

⁸³Surprisingly, the damping time does not depend on the line width of the transition Γ , but narrow transitions lead to a small velocity capture range.

$L_0 = 0.25$ cm for constant deceleration at half the maximum value. Use this simple model of a trap to calculate the capture velocity for rubidium atoms. What is a suitable value for the gradient of the magnetic field, B_0/L (where $L = 2L_0$)? (Data are given in Table 9.1.)

Comment. The magnetic field gradients in a magneto-optical trap are much less than those in magnetic traps (Chapter 10), but the force (from the radiation) is much stronger than the magnetic force.

(9.11) *The equilibrium number of atoms in an MOT*

The steady-state number of atoms that congregate at the centre of an MOT is determined by the balance between the loading rate and the loss caused by collisions. To estimate this equilibrium number $\mathcal{N}$, we consider the trapping region formed at the intersection of the six laser beams of diameter D as being approximately a cube with sides of length D . This trapping region is situated in a cell filled with a low-pressure vapour of number density N .

- (a) The loading rate can be estimated from the kinetic theory expression $\frac{1}{4}NvAf(v)$ for the rate at which atoms with speed v hit a surface of area A in a gas; $f(v)$ is the fraction of atoms with speeds in the range v to $v + dv$ (eqn 8.3). Integrate this rate from $v = 0$ up to the capture velocity v_c to obtain an expression for the rate at which the MOT captures atoms from the vapour. (The integration can be made simple by assuming that $v_c \ll v_p$.)
- (b) Atoms are lost ('knocked out of the trap') by collisions with fast atoms in the vapour at a rate $\dot{\mathcal{N}} = -N\bar{v}\sigma$, where $\bar{v}$ is the mean velocity in the vapour and σ is a collision cross-section.

Show that the equilibrium number of atoms in the MOT is independent of the vapour pressure.

- (c) Atoms enter the trapping region over a surface area $A = 6D^2$. An MOT with $D = 2$ cm has $v_c \simeq 25 \text{ m s}^{-1}$ for rubidium. Make a reasonable estimate of the cross-section σ for collisions between two atoms and hence find the equilibrium number of atoms captured from a low-pressure vapour at room temperature.⁸⁴

(9.12) *Optical absorption by cold atoms in an MOT*

In a simplified model the trapped atoms are considered as a spherical cloud of uniform density, radius r and number $\mathcal{N}$.

- (a) Show that a laser beam of (angular) frequency ω that passes through the cloud has a fractional change in intensity given by

$$\frac{\Delta I}{I_0} \simeq \frac{\mathcal{N}\sigma(\omega)}{2r^2}.$$

(The optical absorption cross-section $\sigma(\omega)$ is given by eqn 7.76.)

- (b) Absorption will significantly affect the operation of the trap when $\Delta I \simeq I_0$. Assuming that this condition limits the density of the cloud for large numbers of atoms,⁸⁵ estimate the radius and density for a cloud of $\mathcal{N} = 10^9$ rubidium atoms and a frequency detuning of $\delta = -2\Gamma$.

(9.13) *Laser cooling of atoms with hyperfine structure*

The treatment of Doppler cooling given in the text assumes a two-level atom, but in real experiments with the optical molasses technique, or the MOT, the hyperfine structure of the ground configuration causes complications. This exercise goes through

⁸⁴The equilibrium number in the trap is independent of the background vapour pressure over a wide range, between (i) the pressure at which collisions with fast atoms knock cold atoms out of the trap before they settle to the centre of the trap, and (ii) the much lower pressure at which the loss rate from collisions between the cold atoms within the trap itself becomes important compared to collisions with the background vapour.

⁸⁵*Comment.* Actually, absorption of the laser beams improves the trapping—for an atom on the edge of the cloud the light pushing outwards has a lower intensity than the unattenuated laser beam directed inwards—however, the spontaneously-emitted photons associated with this absorption do cause a problem when the cloud is 'optically thick', i.e. when most of the light is absorbed. Under these conditions a photon emitted by an atom near the centre of the trap is likely to be reabsorbed by another atom on its way out of the cloud—this scattering within the cloud leads to an outward radiation pressure (similar to that in stars) that counteracts the trapping force of the six laser beams. (The additional scattering also increases the rate of heating for a given intensity of the light.) In real experiments, however, the trapping force in the MOT is not spherically symmetric, and there may be misalignments or other imperfections of the laser beams, so that for conditions of high absorption the cloud of trapped atoms tends to become unstable (and may spill out of the trap).

some of the nitty-gritty details and tests understanding of hyperfine structure.⁸⁶

- (a) Sodium has a nuclear spin $I = 3/2$. Draw an energy level diagram of the hyperfine structure of the $3s\ ^2S_{1/2}$ and $3p\ ^2P_{3/2}$ levels and indicate the allowed electric dipole transitions.
- (b) In a laser cooling experiment the transition $3s\ ^2S_{1/2}$, $F = 2$ to $3p\ ^2P_{3/2}$, $F' = 3$ is excited by light that has a frequency detuning of $\delta = -\Gamma/2 \simeq -5\text{ MHz}$ (to the red of this transition). Selection rules dictate that the excited state decays back to the initial state, so there is a nearly closed cycle of absorption and spontaneous emission, but there is some off-resonant excitation to the $F' = 2$ hyperfine level which can decay to $F = 1$ and be 'lost' from the cycle. The $F' = 2$ level lies 60 MHz below the $F' = 3$ level. Estimate the average number of photons scattered by an atom before it falls into the lower hyperfine level of the ground configuration. (Assume that the transitions have similar strengths.)
- (c) To counteract the leakage out of the laser cooling cycle, experiments use an additional laser beam that excites atoms out of the $3s\ ^2S_{1/2}$, $F = 1$ level (so that they can get back into the $3s\ ^2S_{1/2}$, $F = 2$ level). Suggest a suitable transition for this 'repumping' process and comment on the light intensity required.⁸⁷

(9.14) *The gradient force*

Figure 9.11 shows a sphere, with a refractive index greater than the surrounding medium, that feels a force towards regions of high intensity. Draw a similar diagram for the case $n_{\text{sphere}} < n_{\text{medium}}$ and indicate forces. (This object could be a small bubble of air in a liquid.)

(9.15) *Dipole-force trap*

A laser beam⁸⁸ propagating along the z -axis has an intensity profile of

$$I = \frac{2P}{\pi w(z)^2} \exp\left(-\frac{2r^2}{w(z)^2}\right), \quad (9.60)$$

where $r^2 = x^2 + y^2$ and the waist size is $w(z) = w_0(1 + z^2/b^2)^{1/2}$ with $b = \pi w_0^2/\lambda^2$. This laser beam has a power of $P = 1\text{ W}$ at a wavelength of $\lambda = 1.06\ \mu\text{m}$, and a spot size of $w_0 = 10\ \mu\text{m}$ at the focus.

- (a) Show that the integral of $I(r, z)$ over any plane of constant z equals the total power of the beam P .
- (b) Calculate the depth of the dipole potential for rubidium atoms, expressing your answer as an equivalent temperature.
- (c) For atoms with a thermal energy much lower than the trap depth (so that $r^2 \ll w_0^2$ and $z^2 \ll b^2$), determine the ratio of the size of the cloud in the radial and longitudinal directions.
- *(d) Show that the dipole force has a maximum value at a radial distance of $r = w_0/2$. Find the maximum value of the waist size w_0 for which the dipole-force trap supports rubidium atoms against gravity (when the laser beam propagates horizontally).

(9.16) *An optical lattice*

In a standing wave of radiation with a wavelength of $\lambda = 1.06\ \mu\text{m}$, a sodium atom experiences a periodic potential as in eqn 9.52 with $U_0 = 100E_r$, where E_r is the recoil energy (for light at the atom's resonance wavelength $\lambda_0 = 0.589\ \mu\text{m}$). Calculate the oscillation frequency for a cold atom trapped near the bottom of a potential well in this one-dimensional optical lattice. What is the energy spacing between the low-lying vibrational levels?

(9.17) *The potential for the dipole force*

Show that the force in eqn 9.43 equals the gradient of the potential

$$U_{\text{dipole}} = -\frac{\hbar\delta}{2} \ln\left(1 + \frac{I}{I_{\text{sat}}} + \frac{4\delta^2}{\Gamma^2}\right).$$

For what conditions does eqn 9.46 give a good approximation for U_{dipole} ?

⁸⁶The transfer between different hyperfine levels described here is distinct from the transfer between different Zeeman sub-levels (states of given M_J or M_F) in the Sisyphus effect.

⁸⁷In the Zeeman slowing technique the magnetic field increases the separation of the energy levels (and also uncouples the nuclear and electronic spins in the excited state) so that 'repumping' is not necessary.

⁸⁸This is a diffraction-limited Gaussian beam.